Decision Trees

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Predicting potential loan defaults

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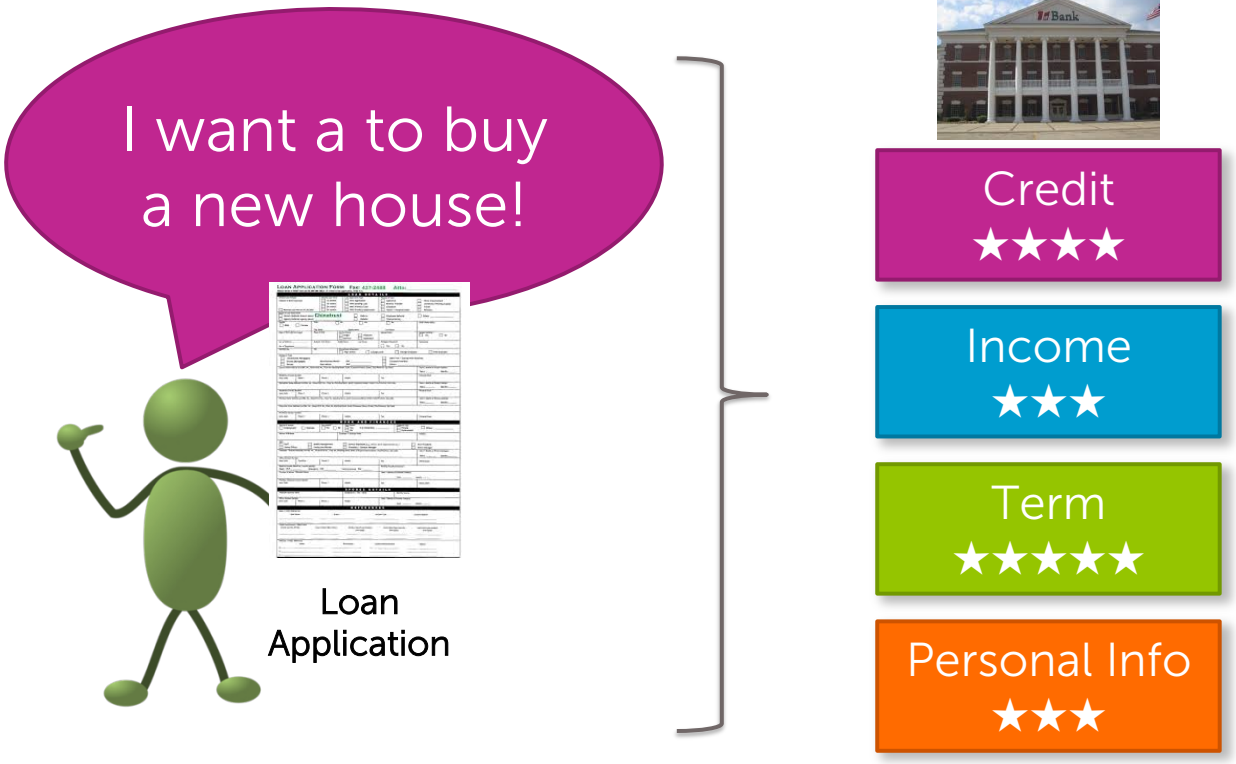
What makes a loan risky?



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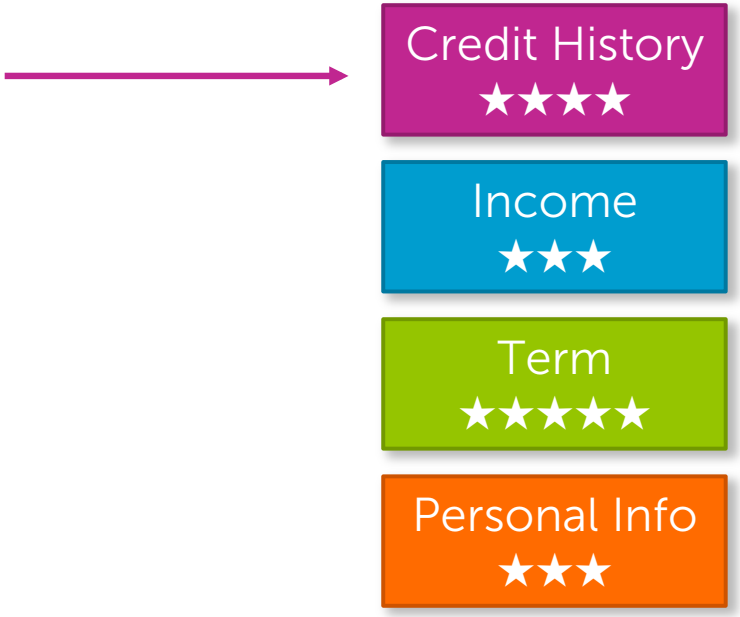
Credit history explained

Did I pay previous loans on time?

Example: excellent, good, or fair



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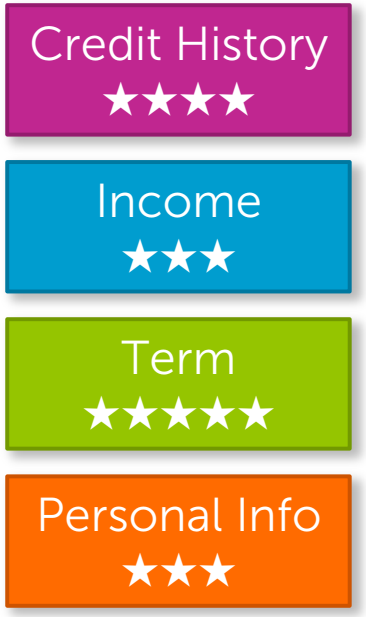
Income

What's my income?

Example:
$80K per year



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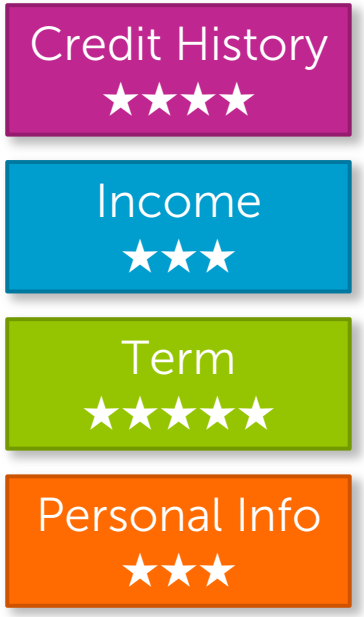
Loan terms

How soon do I need to pay the loan?

Example: 3 years, 5 years,…



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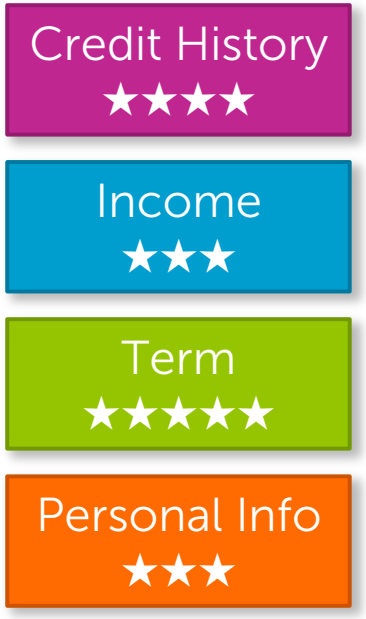
Personal information

Age, reason for the loan, marital status,…

Example: Home loan for a married couple



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Intelligent application

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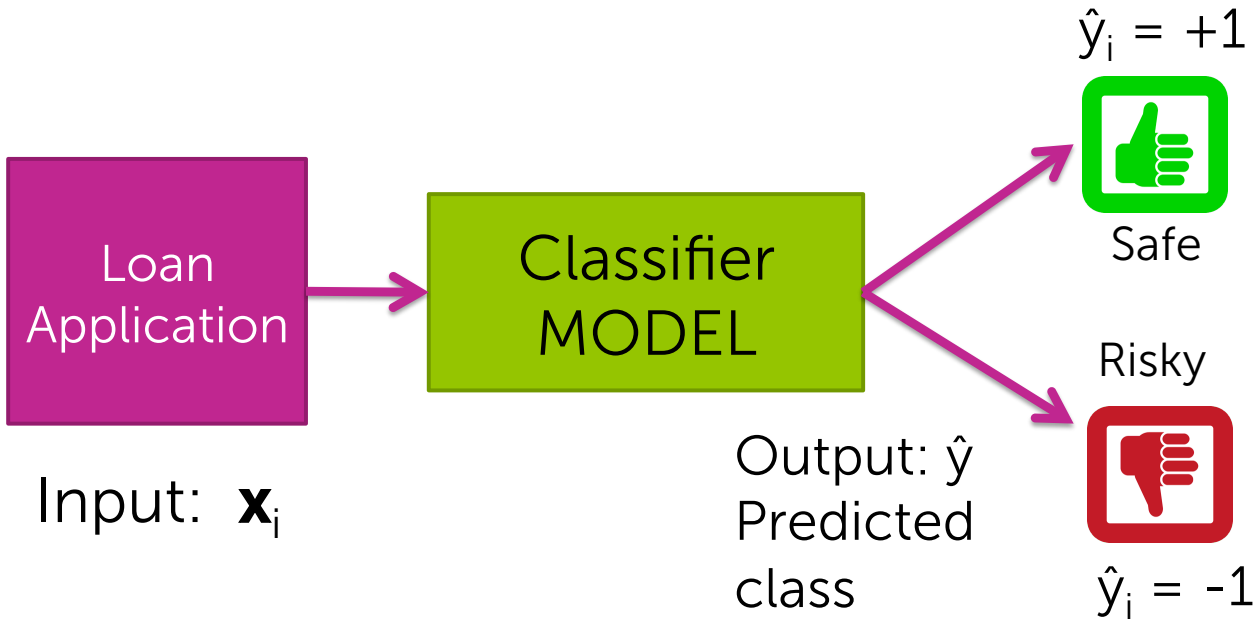
Classifier review



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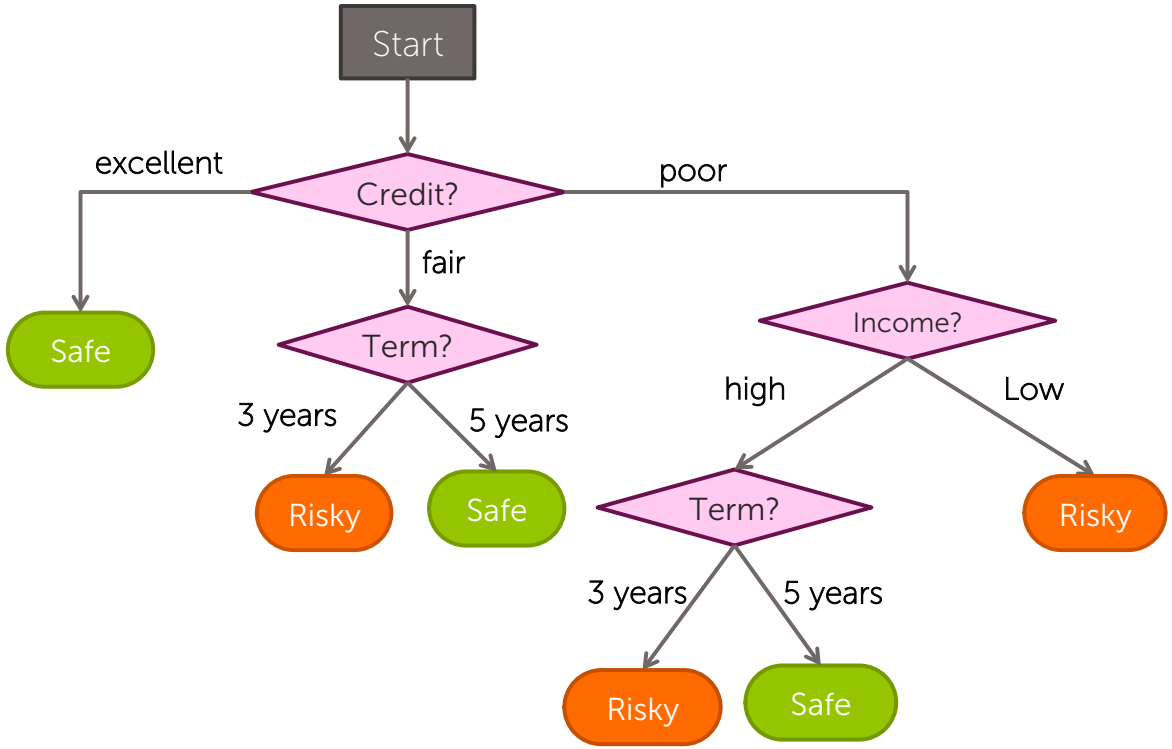
This module … decision trees



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Decision trees: *Intuition*

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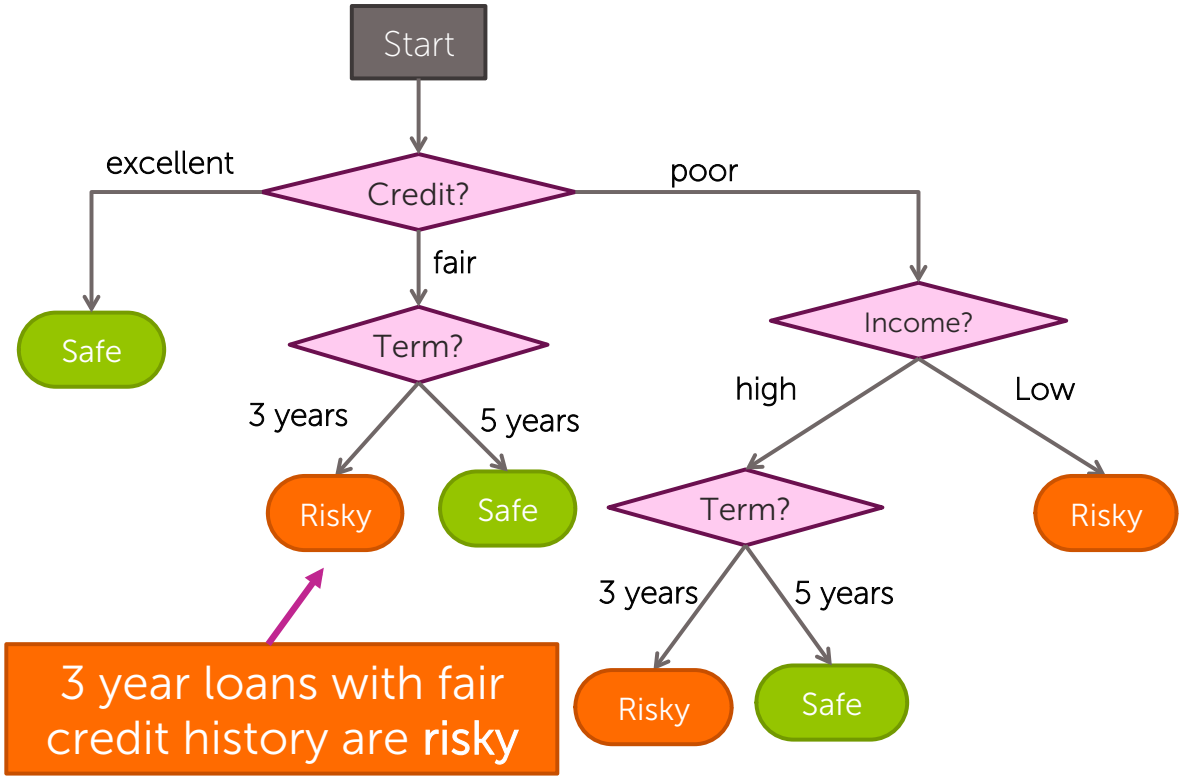
What does a decision tree represent?



3 year loans with fair credit history are **risky**

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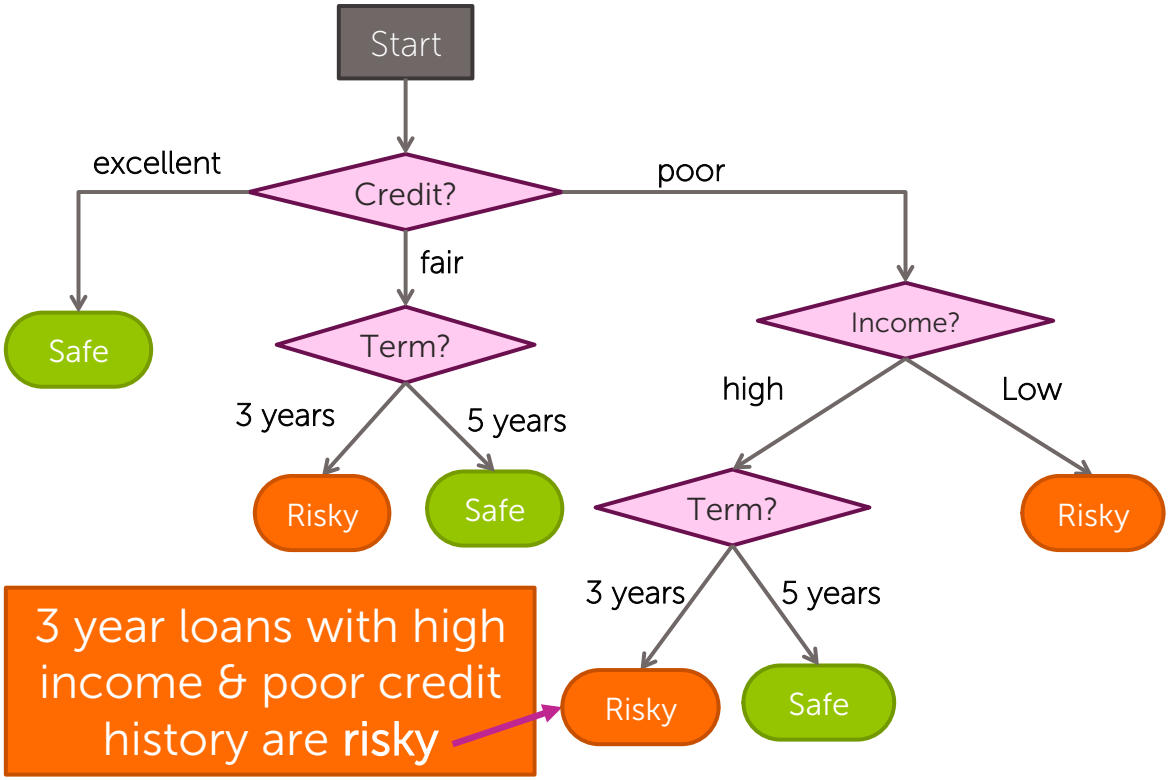
What does a decision tree represent?



3 year loans with high income & poor credit history are **risky**

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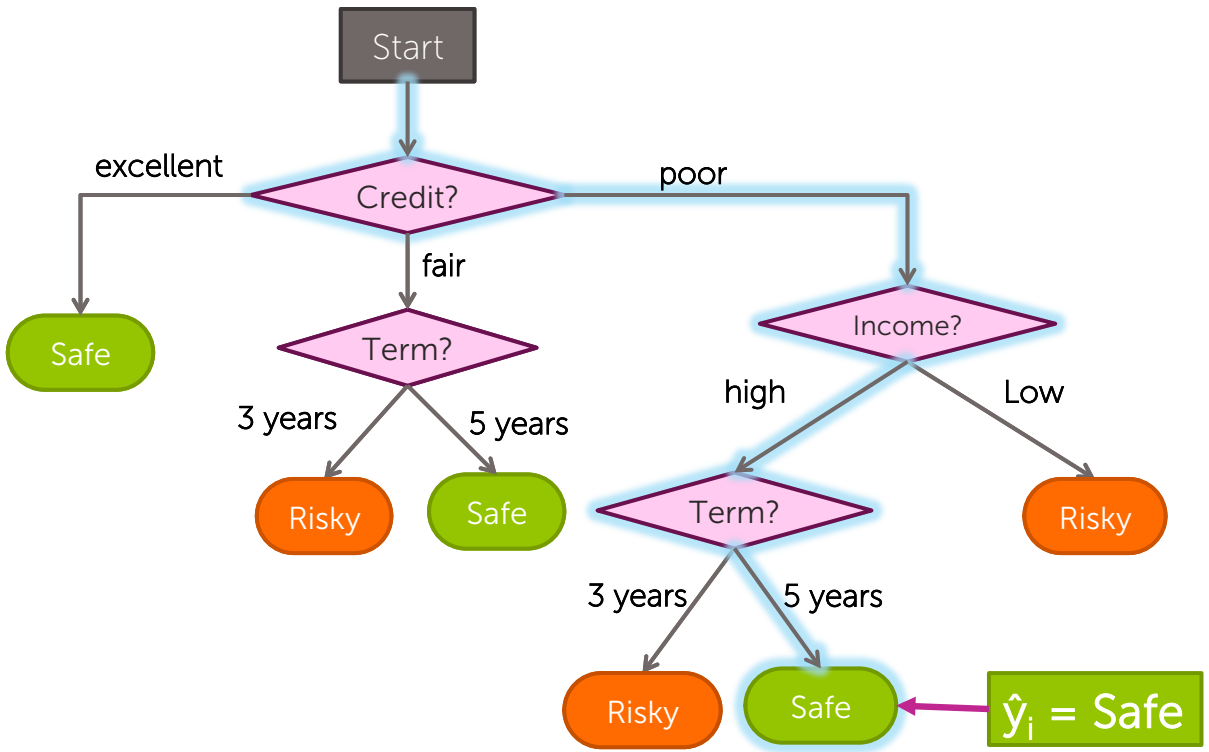
Scoring a loan application

$\mathbf{x}_i$ = (Credit = poor, Income = high, Term = 5 years)



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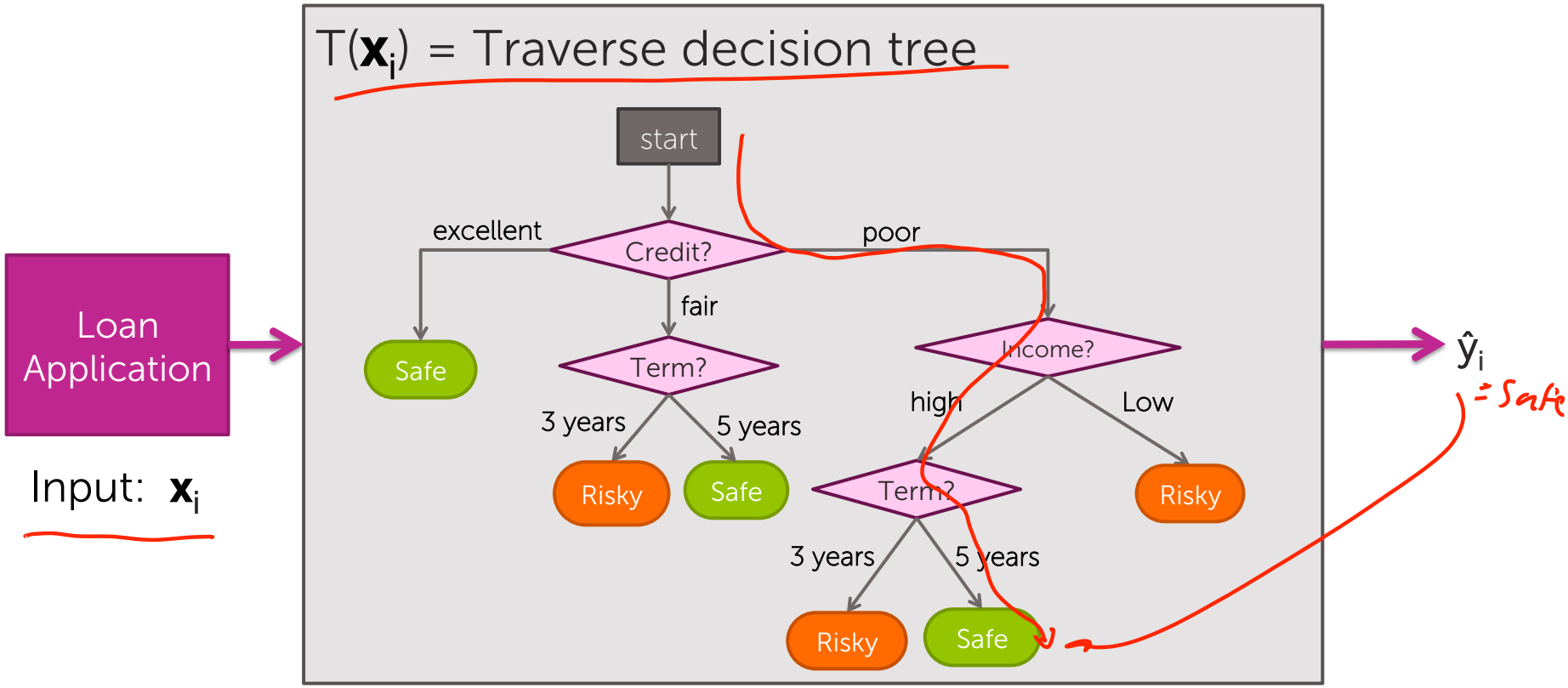
Decision tree model



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Decision tree learning task

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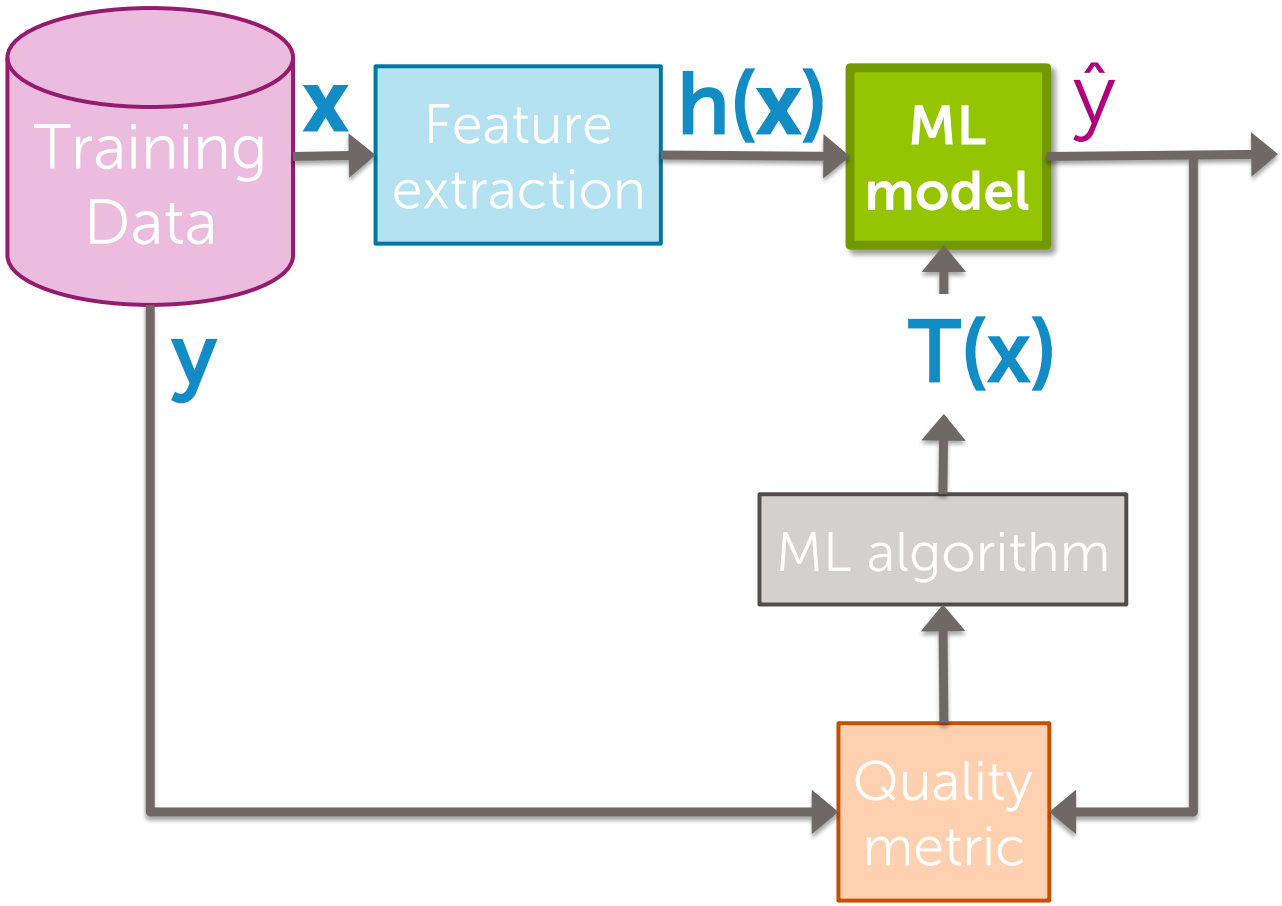

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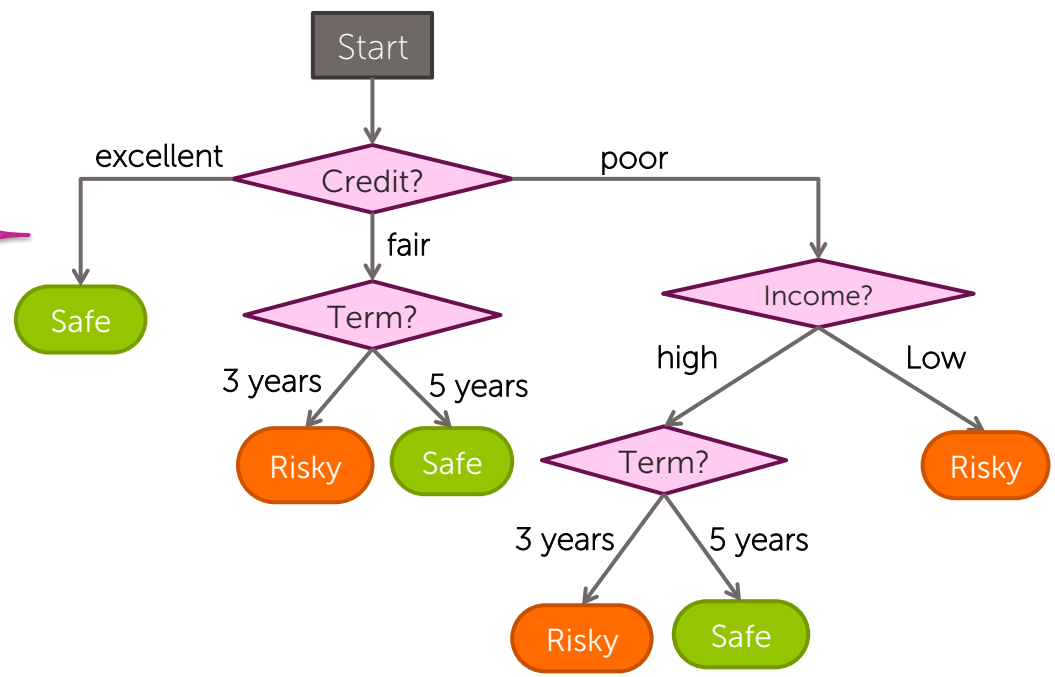
Learn decision tree from data?

$h_1(x)$	$h_2(x)$	$h_3(x)$	Loan Status
Credit	**Term**	**Income**	**y**
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe



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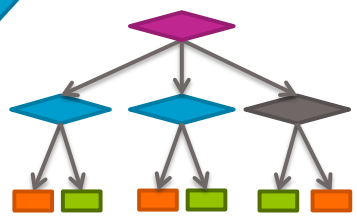
Decision tree learning problem

Training data: N observations ($\mathbf{x}_i$,y_i)

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe

Optimize **quality metric** on training data

T(X)



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Quality metric: Classification error

- Error measures fraction of mistakes

$$\text{Error} = \frac{\text{\# incorrect predictions}}{\text{\# examples}}$$

  - Best possible value : 0.0
  - Worst possible value: 1.0

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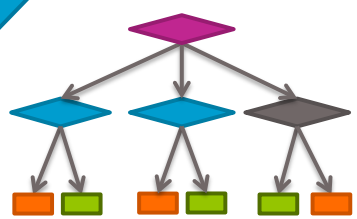
Find the tree with lowest classification error

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe

Minimize **classification error** on training data

T(X)



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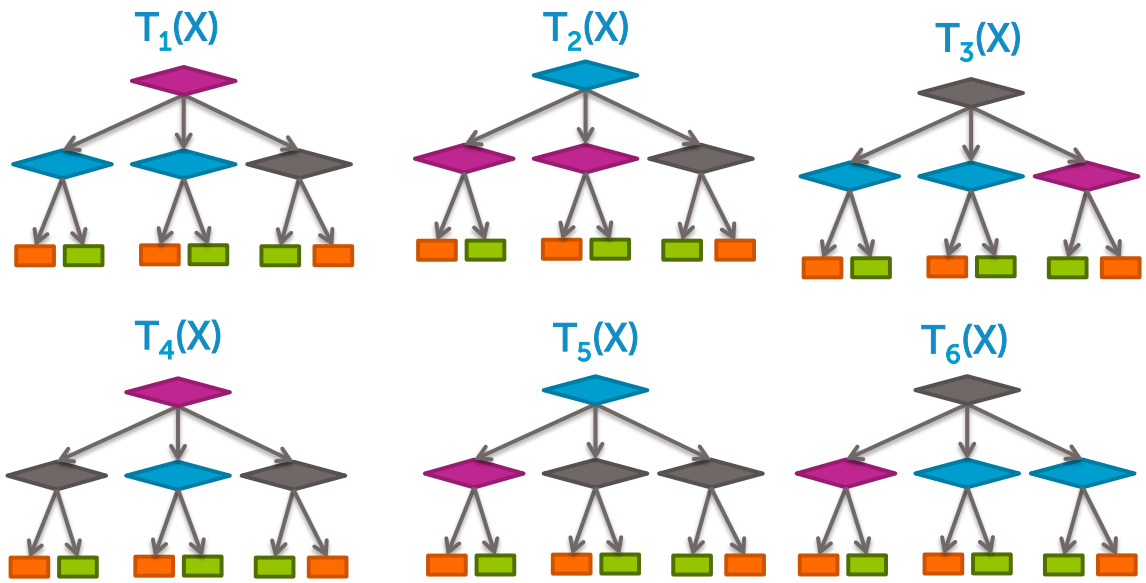
How do we find the best tree?

Exponentially large number of possible trees makes decision tree learning hard!
(NP-hard problem)



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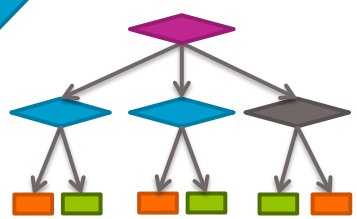
Simple (greedy) algorithm finds "good" tree

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe

Approximately minimize **classification error** on training data

T(X)



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Greedy decision tree learning:
Algorithm outline

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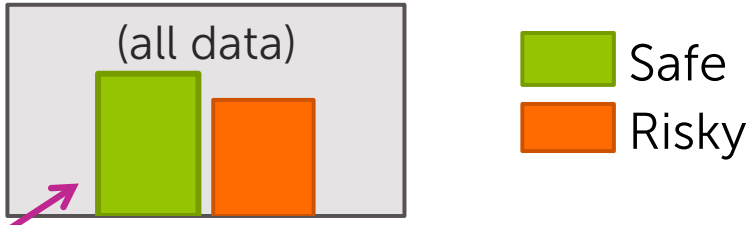
Step 1: Start with an empty tree



Histogram of y values

All points in the dataset

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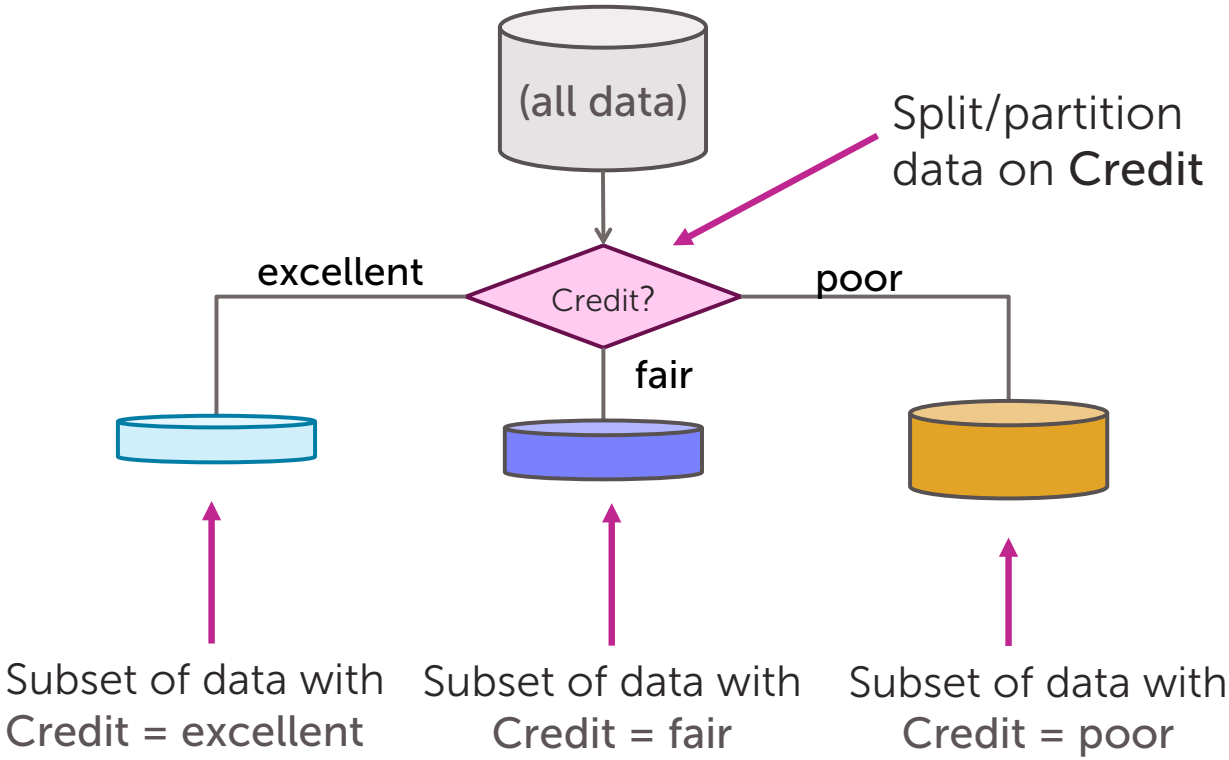
Step 2: Split on a feature



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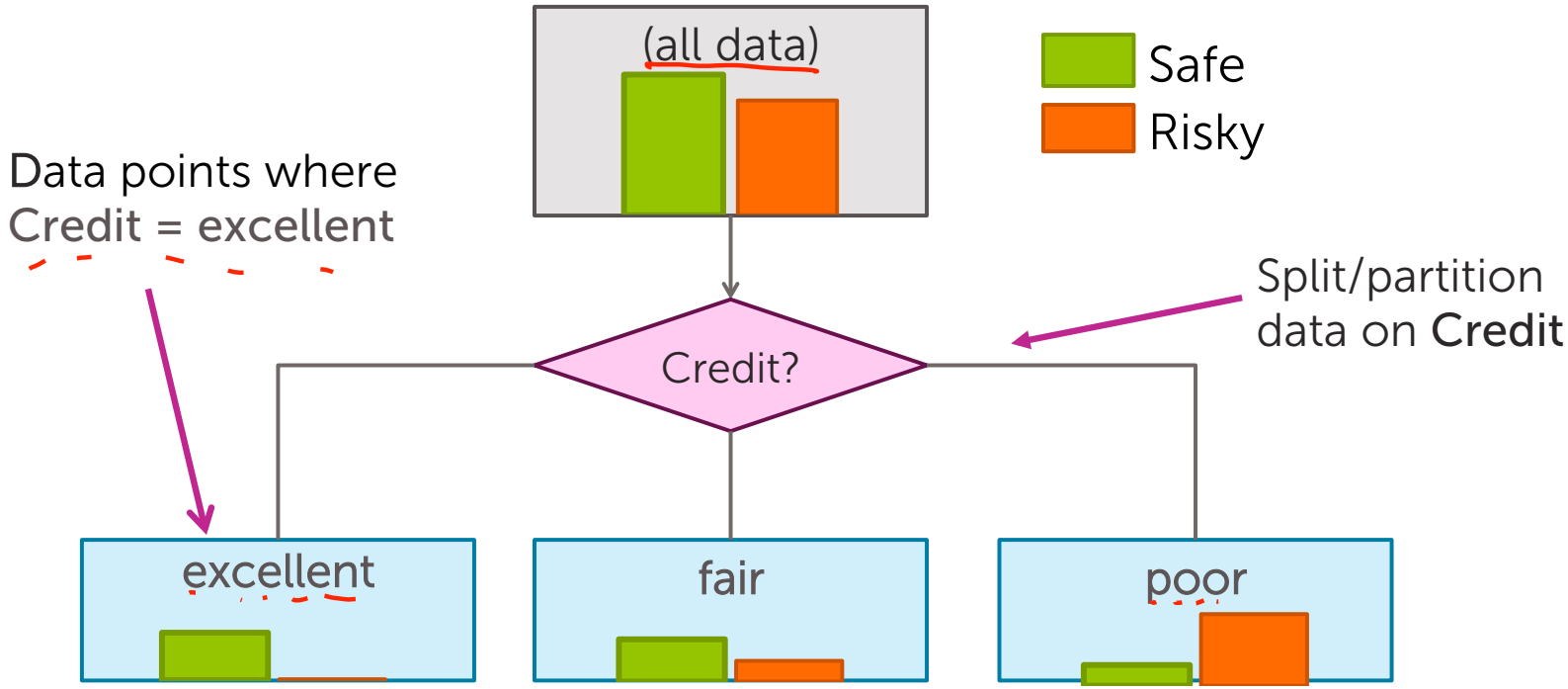
Feature split explained



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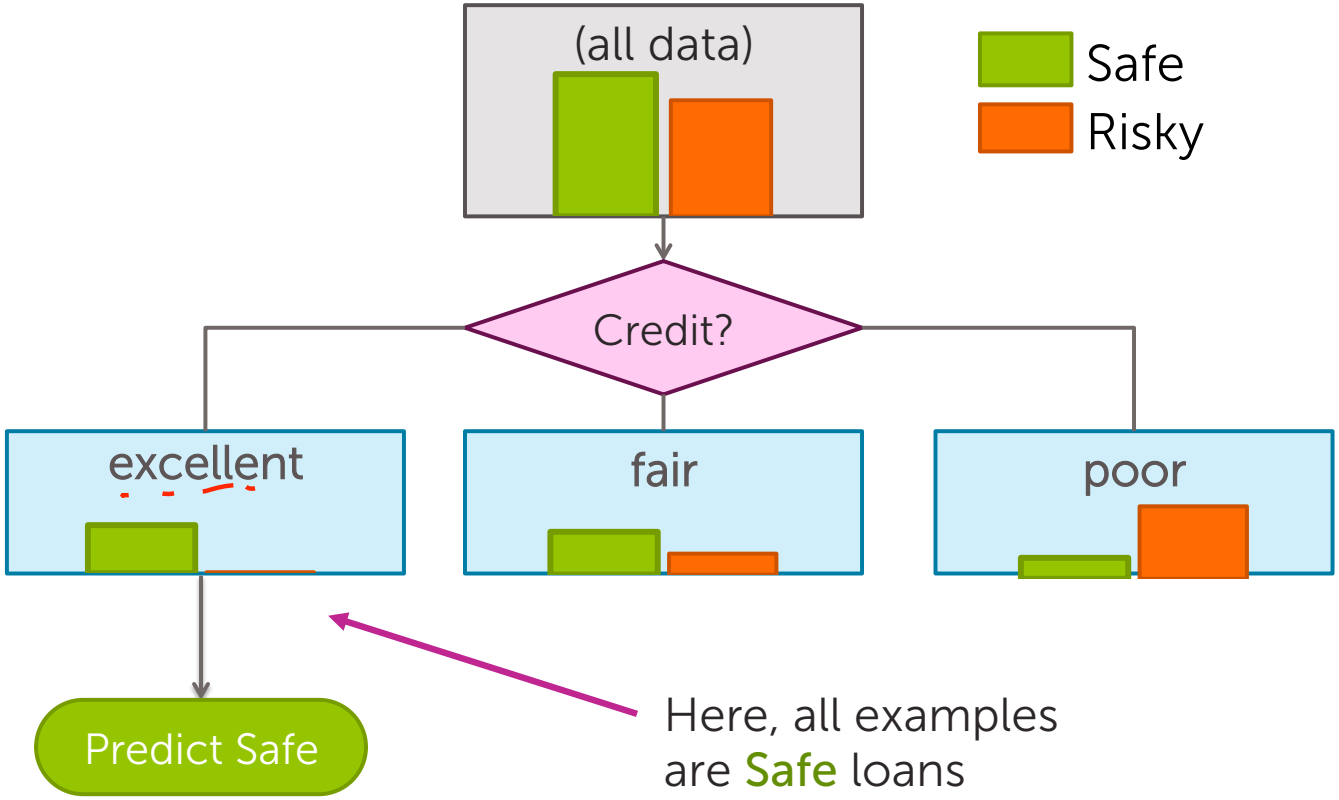
Step 3: Making predictions



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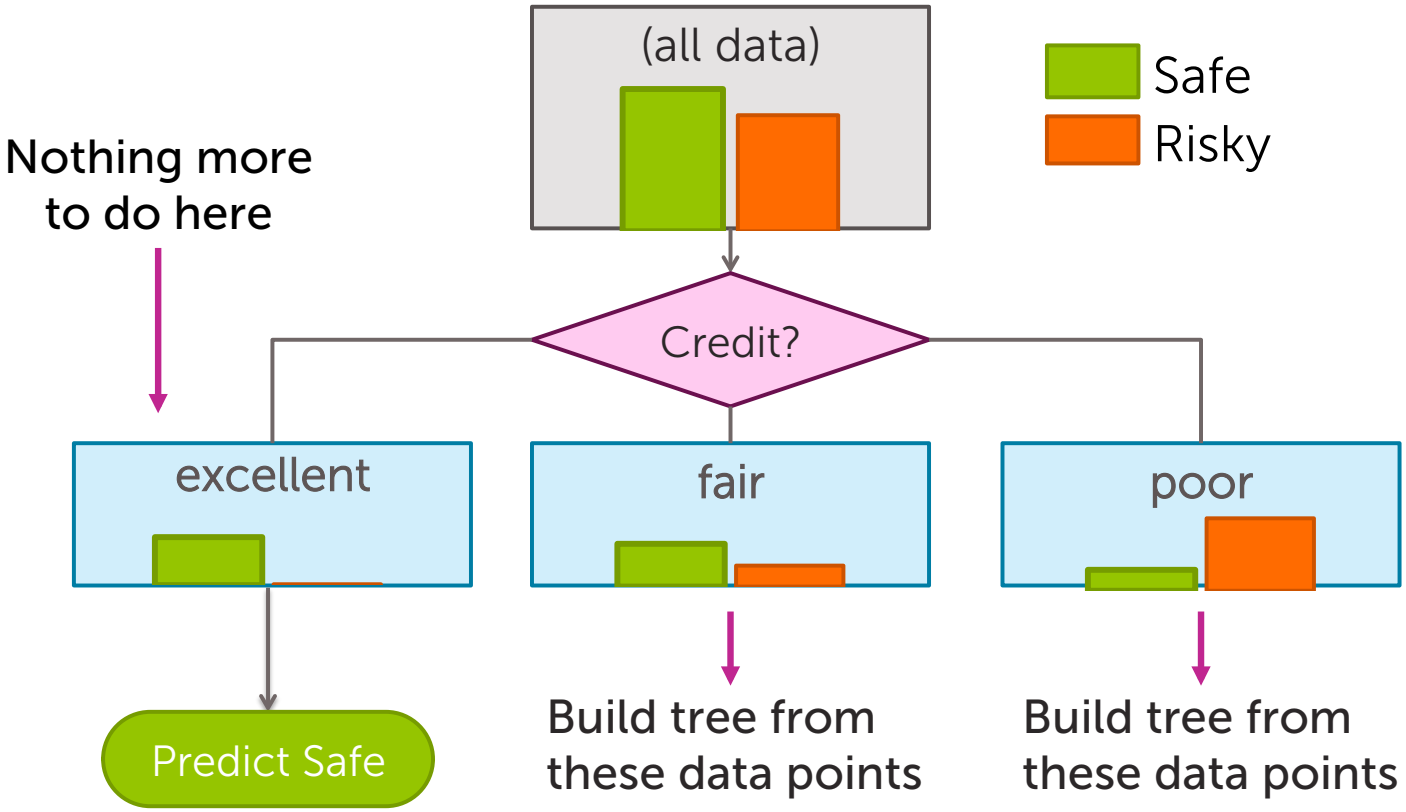
Step 4: Recursion



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Greedy decision tree learning

- Step 1: Start with an empty tree
- Step 2: Select a feature to split data
- For each split of the tree:
  - Step 3: If nothing more to, make predictions
  - Step 4: Otherwise, go to Step 2 & continue (recurse) on this split

Problem 1: Feature split selection

Problem 2: Stopping condition

Recursion

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Feature split learning

=

Decision stump learning

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Start with the data

Assume N = 40, 3 features

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe

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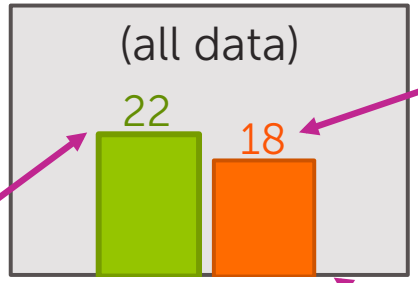
Start with all the data

Loan status: Safe Risky



Number of Risky loans

Number of Safe loans

N = 40 examples

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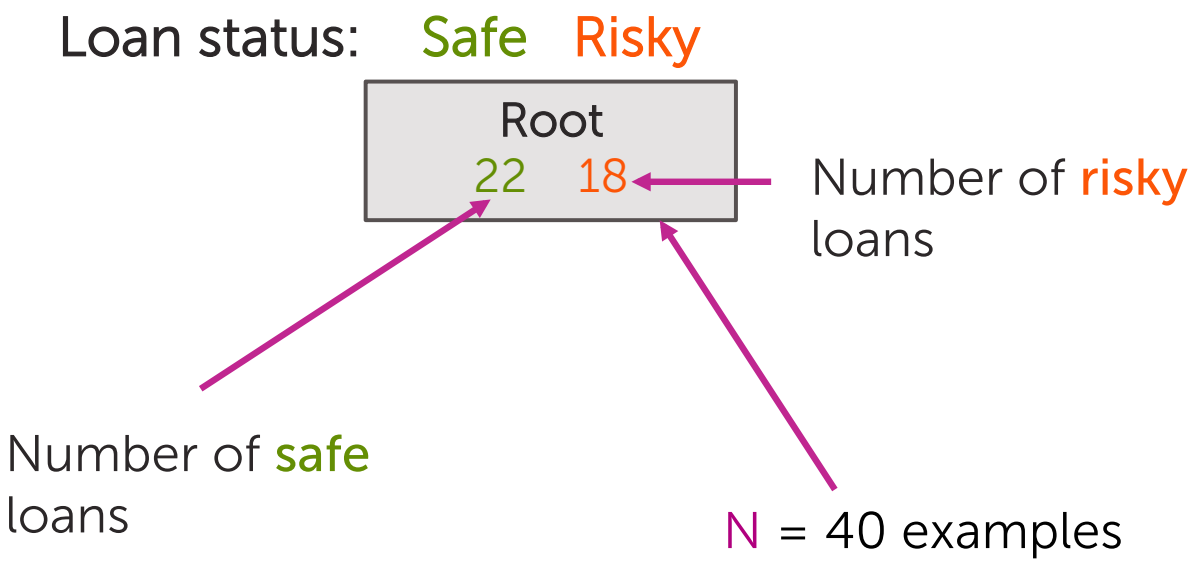
Compact visual notation: Root node



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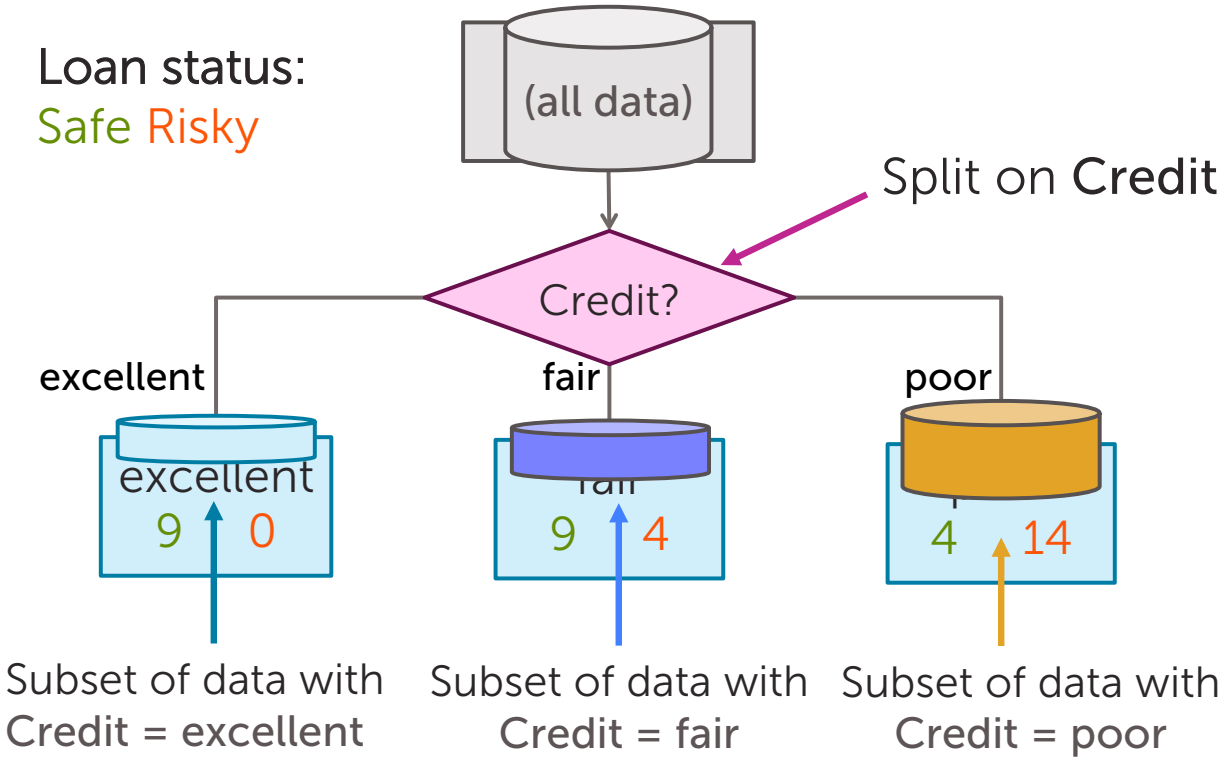
Decision stump: Single level tree



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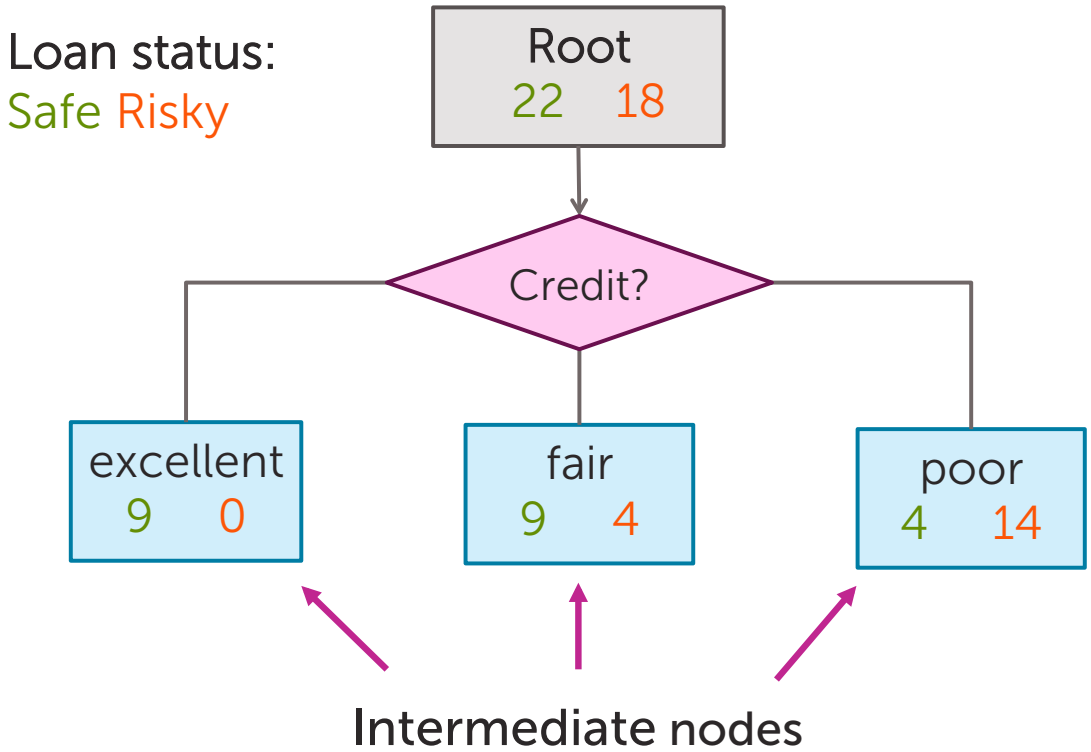
Visual Notation: Intermediate nodes



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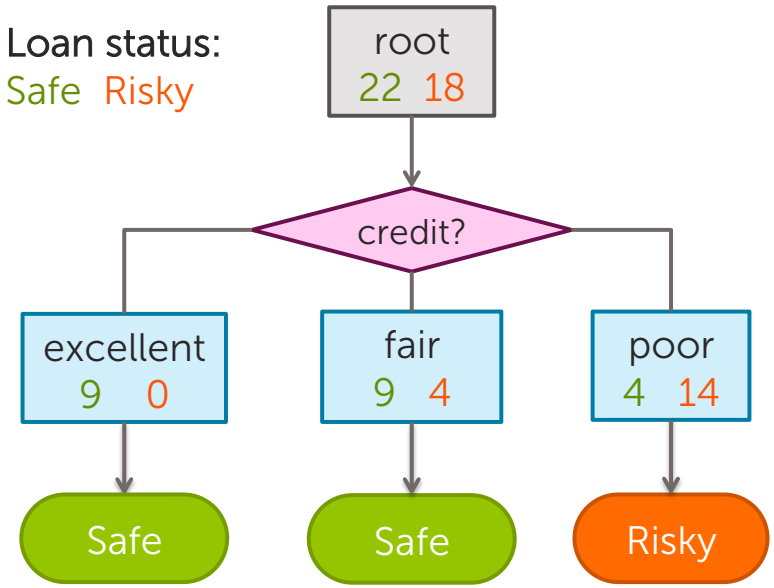
Making predictions with a decision stump



For each intermediate node, set $\hat{y}$ = **majority value**

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Selecting best feature to split on

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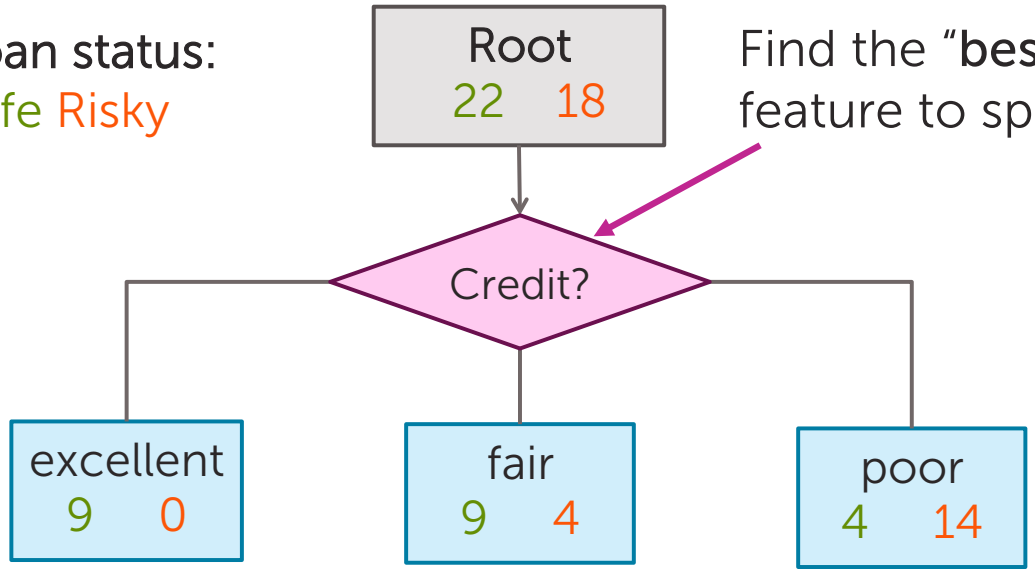
How do we learn a decision stump?

Loan status:
Safe Risky



Find the "**best**" feature to split on!

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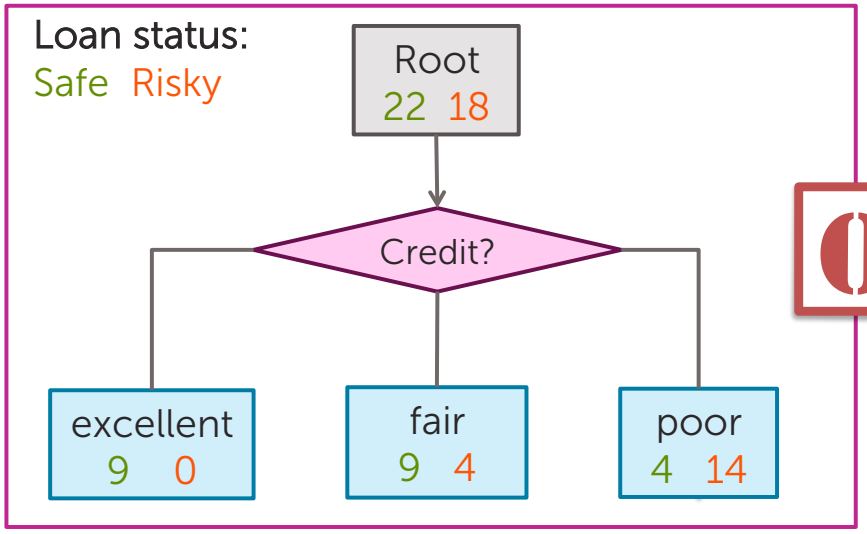
How do we select the best feature?

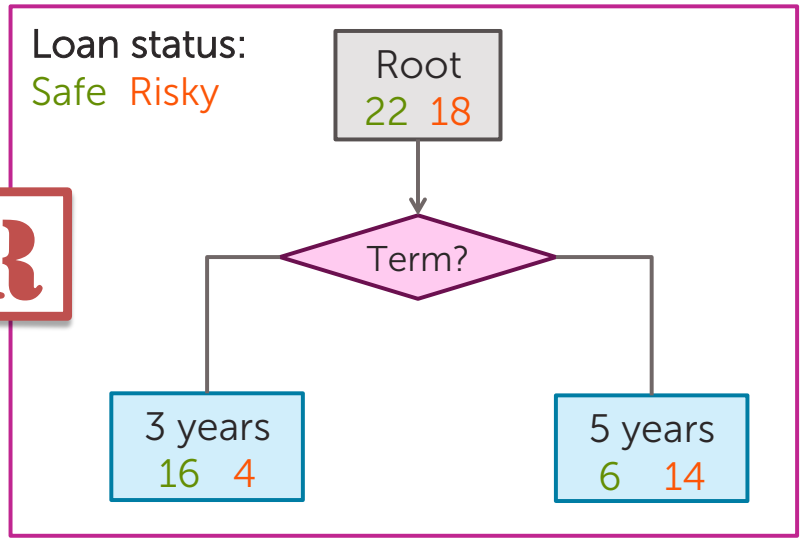
Better?

Choice 1: Split on **Credit**



OR

Choice 2: Split on **Term**



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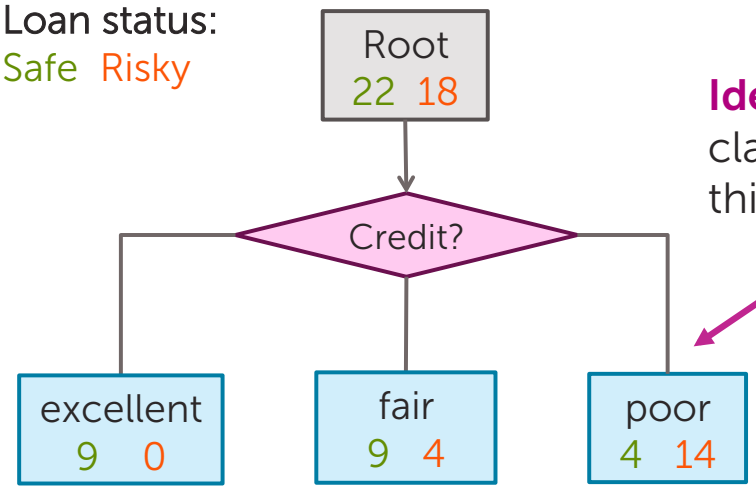
How do we measure effectiveness of a split?

Loan status:
Safe Risky



Idea: Calculate classification error of this decision stump

$$\text{Error} = \frac{\#\text{ mistakes}}{\#\text{ data points}}$$

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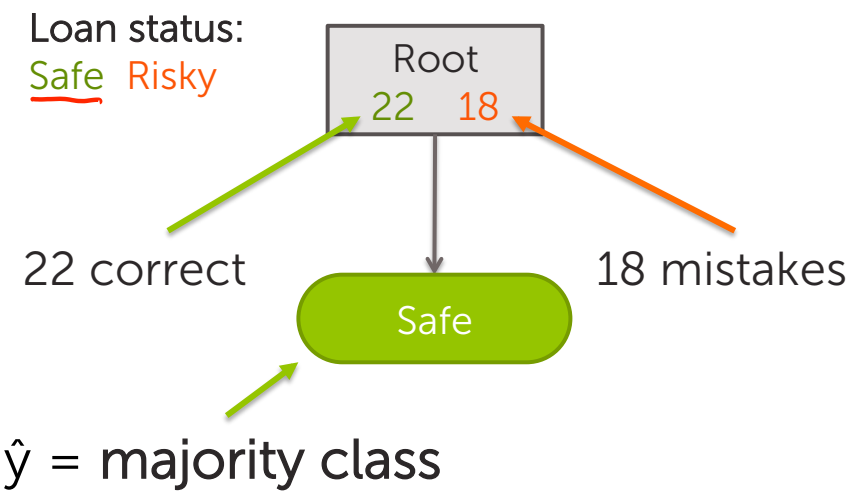
Calculating classification error

- **Step 1:** ŷ = class of majority of data in node
- **Step 2:** Calculate classification error of predicting ŷ for this data



$$\text{Error} = \frac{18}{22+18} = 0.45$$

Tree	Classification error
(root)	0.45

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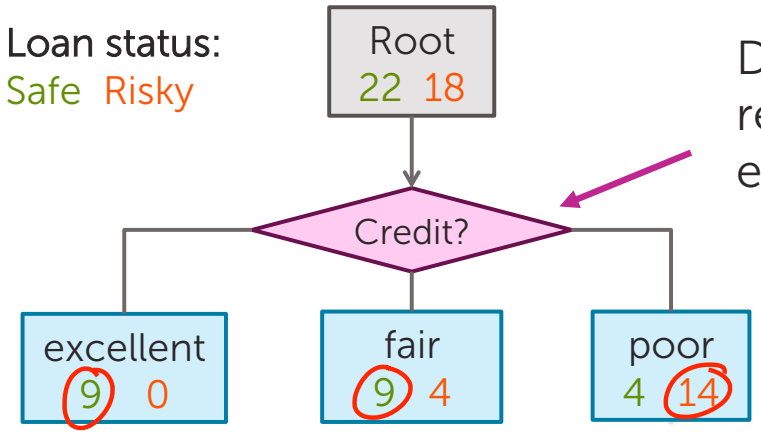
Choice 1: Split on **credit history**?

Choice 1: Split on **Credit**



Does a **split on Credit** reduce classification error below 0.45?

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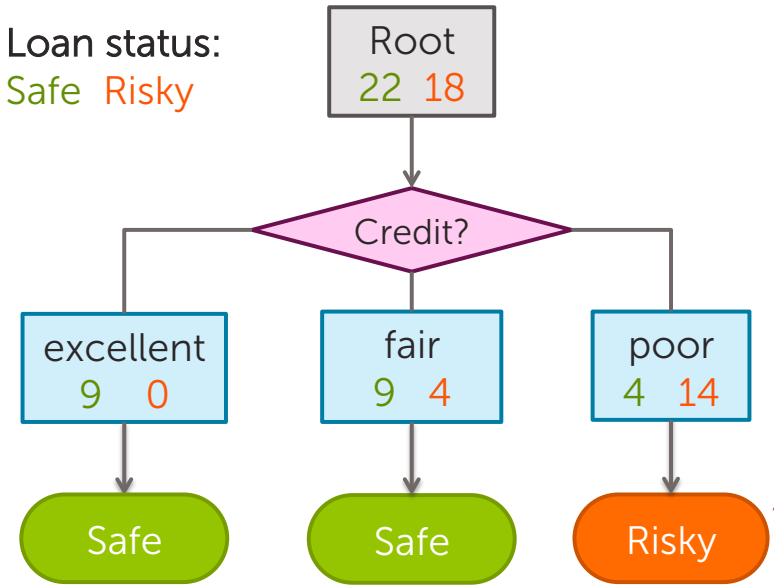
How good is the split on **Credit**?

Choice 1: Split on **Credit**



Step 1: For each intermediate node, set $\hat{y}$ = **majority value**

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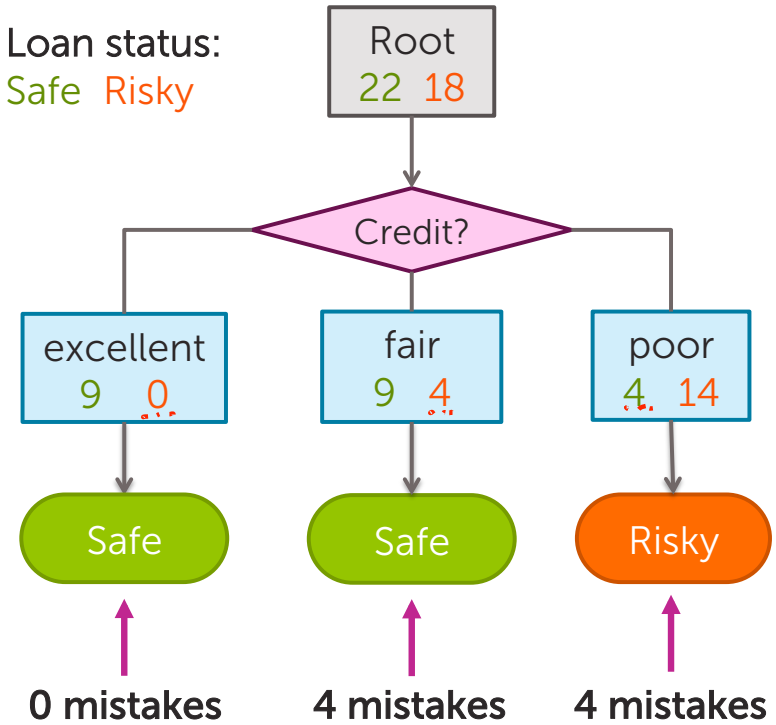
Split on **Credit**: Classification error

Choice 1: Split on **Credit**



$$\text{Error} = \frac{4+4}{40} = 0.20$$

Tree	Classification error
(root)	0.45
Split on **credit**	0.2

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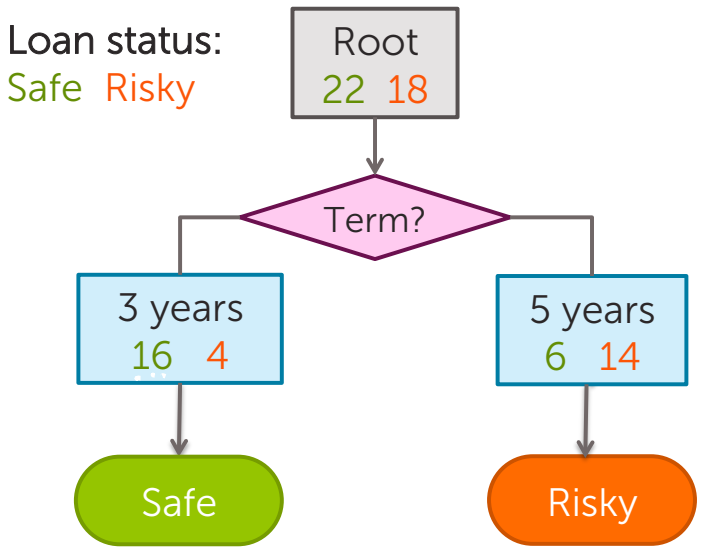
Choice 2: Split on **Term**?

Choice 2: Split on **Term**



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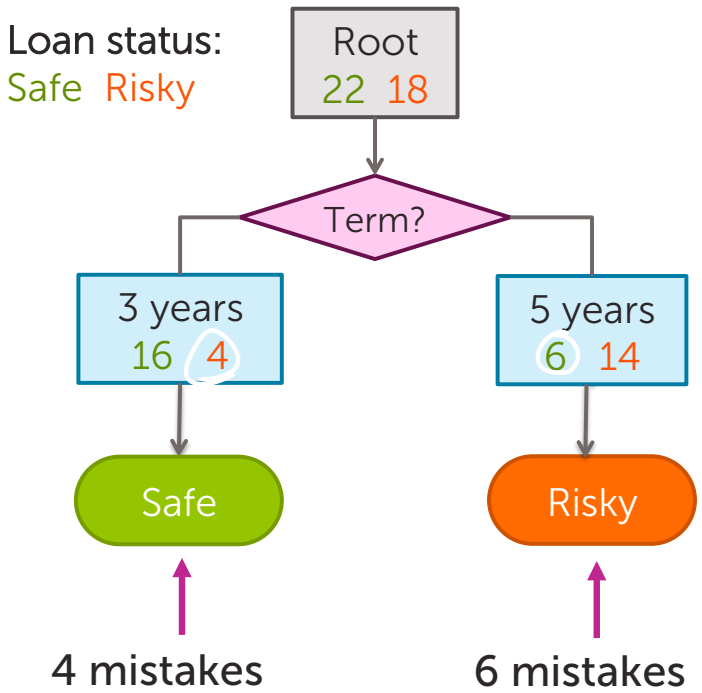
Evaluating the split on **Term**

Choice 2: Split on **Term**



$$\text{Error} = \frac{4+6}{40} = 0.25$$

Tree	Classification error
(root)	0.45
Split on **credit**	0.2
Split on **term**	0.25

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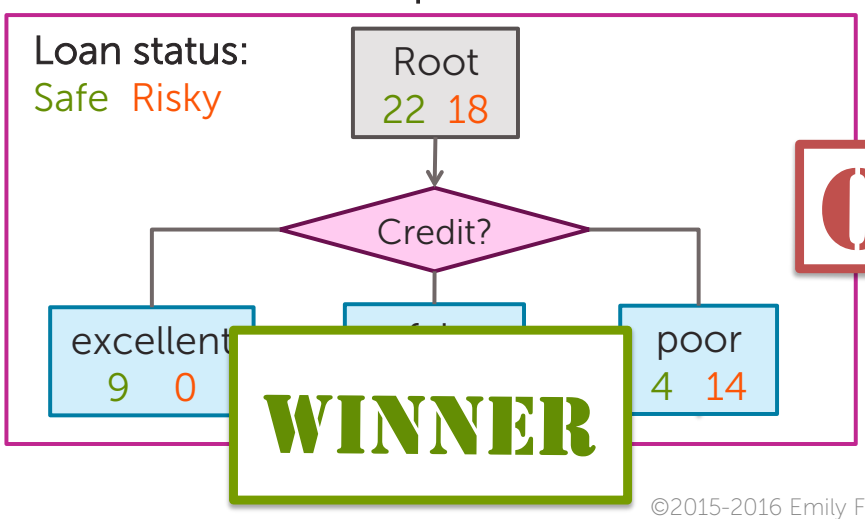
Choice 1 vs Choice 2

Tree	Classification error
(root)	0.45
split on **credit**	0.2
split on **loan term**	0.25

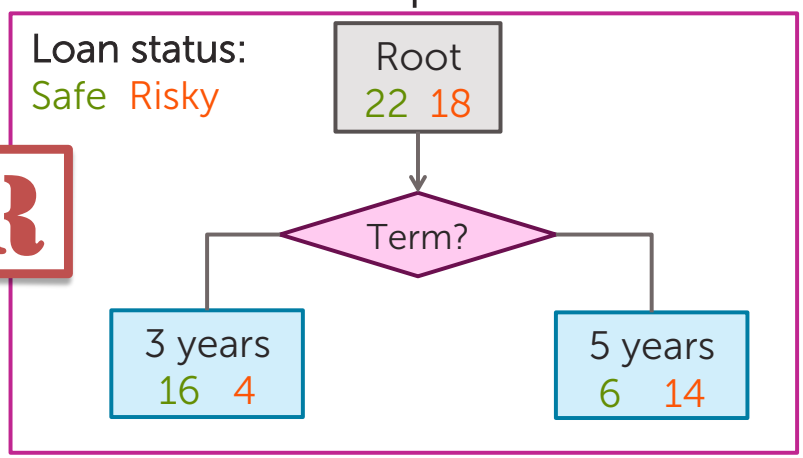
First split!

Choice 1: Split on Credit



OR

Choice 2: Split on Term



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Feature split selection algorithm

- Given a subset of data M (a node in a tree)
- For each feature $h_i(x)$: credit, term, income
  1. Split data of M according to feature $h_i(x)$
  2. Compute classification error split
- Chose feature $h^*(x)$ with lowest classification error credit

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Greedy decision tree learning

- Step 1: Start with an empty tree
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Pick feature split leading to lowest classification error

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Decision Tree Learning:
Recursion & Stopping conditions

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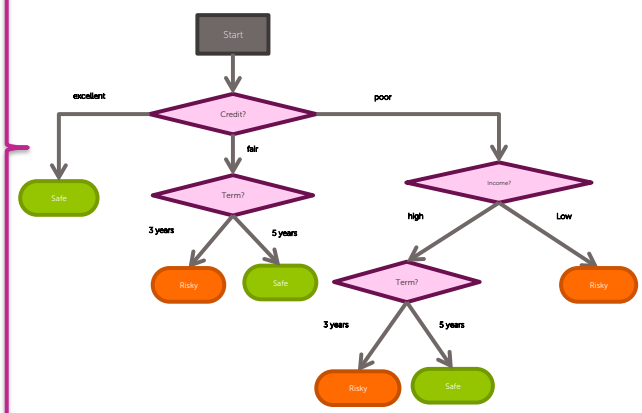
Learn decision tree from data?

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe



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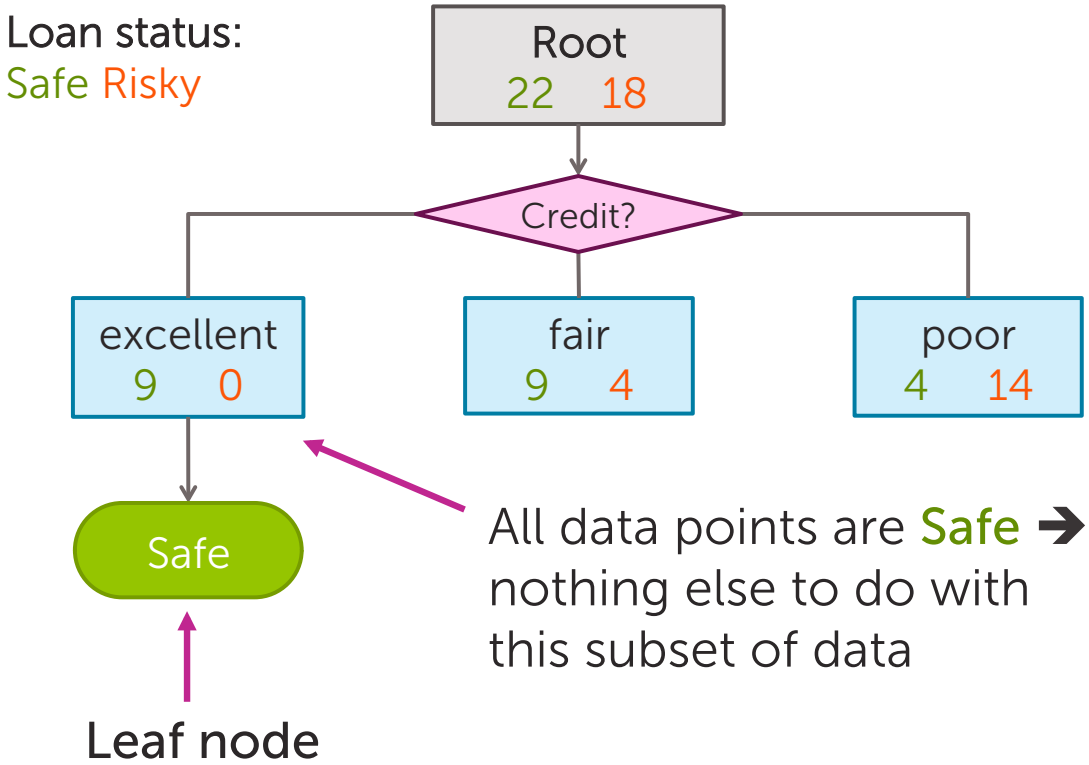
We've learned a decision stump, what next?



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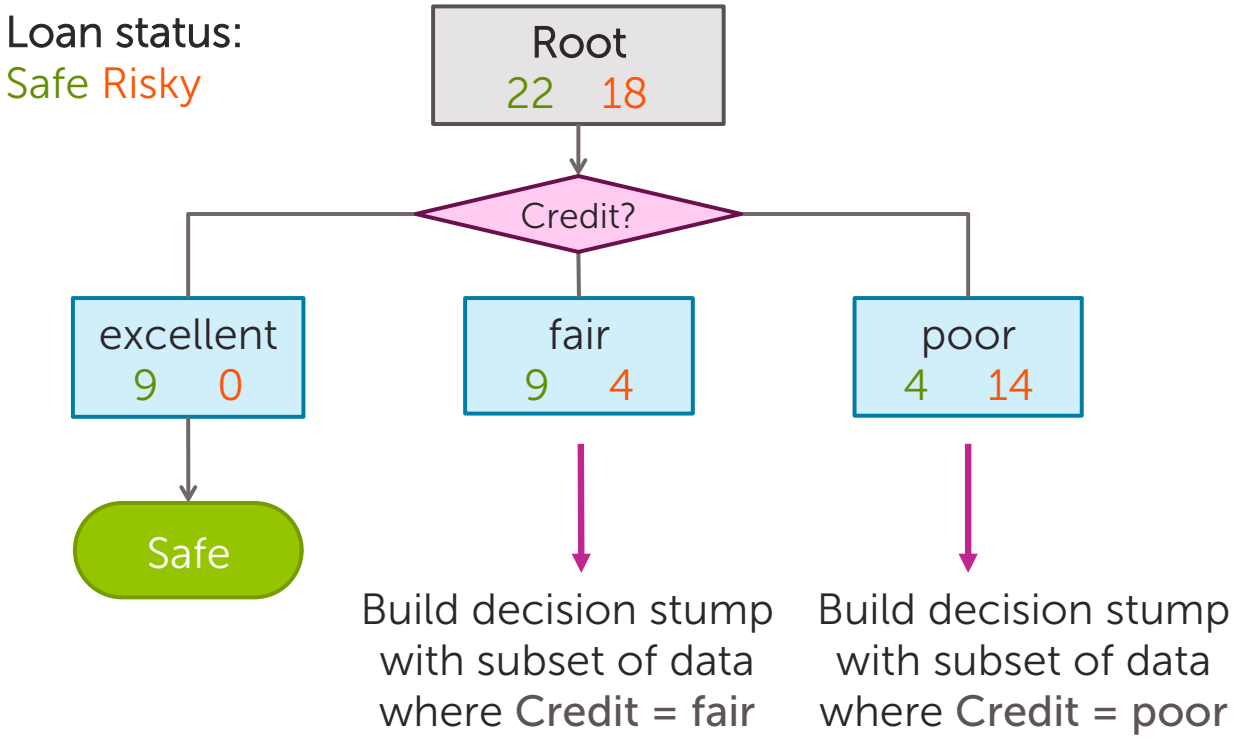
Tree learning = Recursive stump learning



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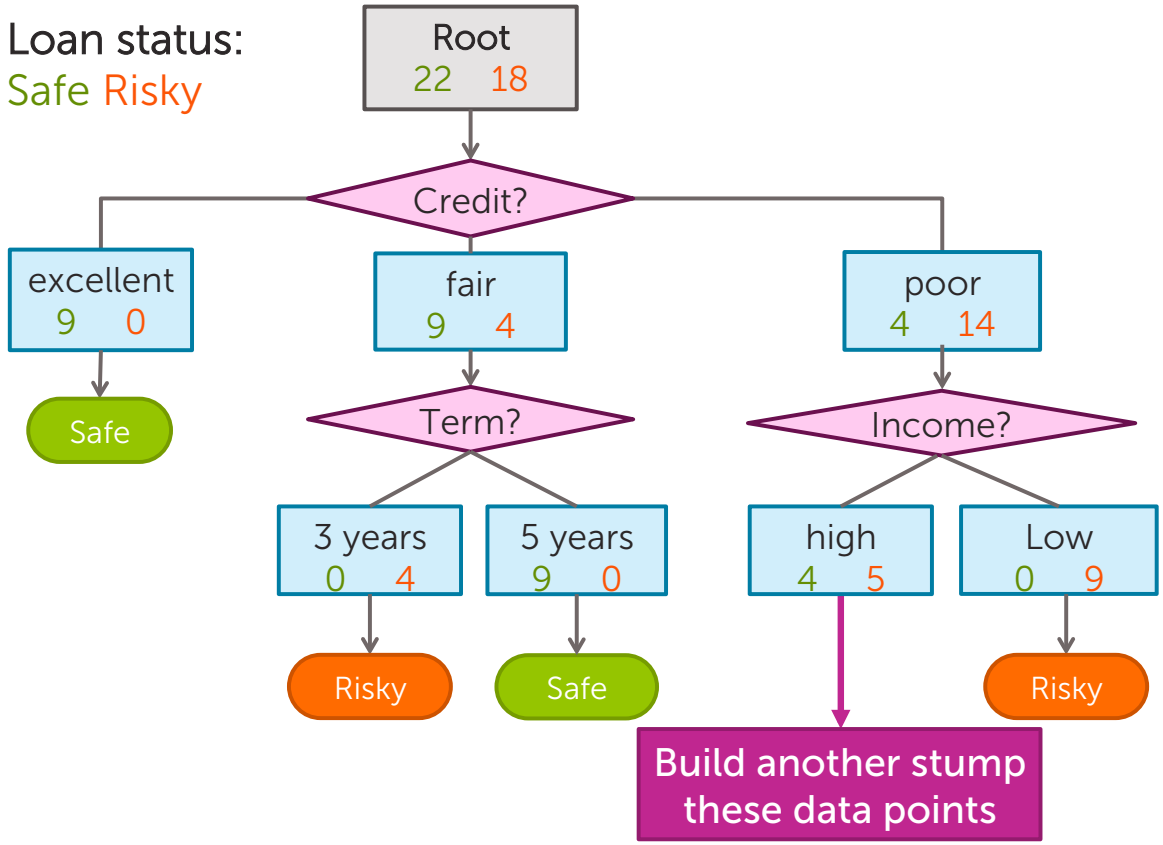
Second level



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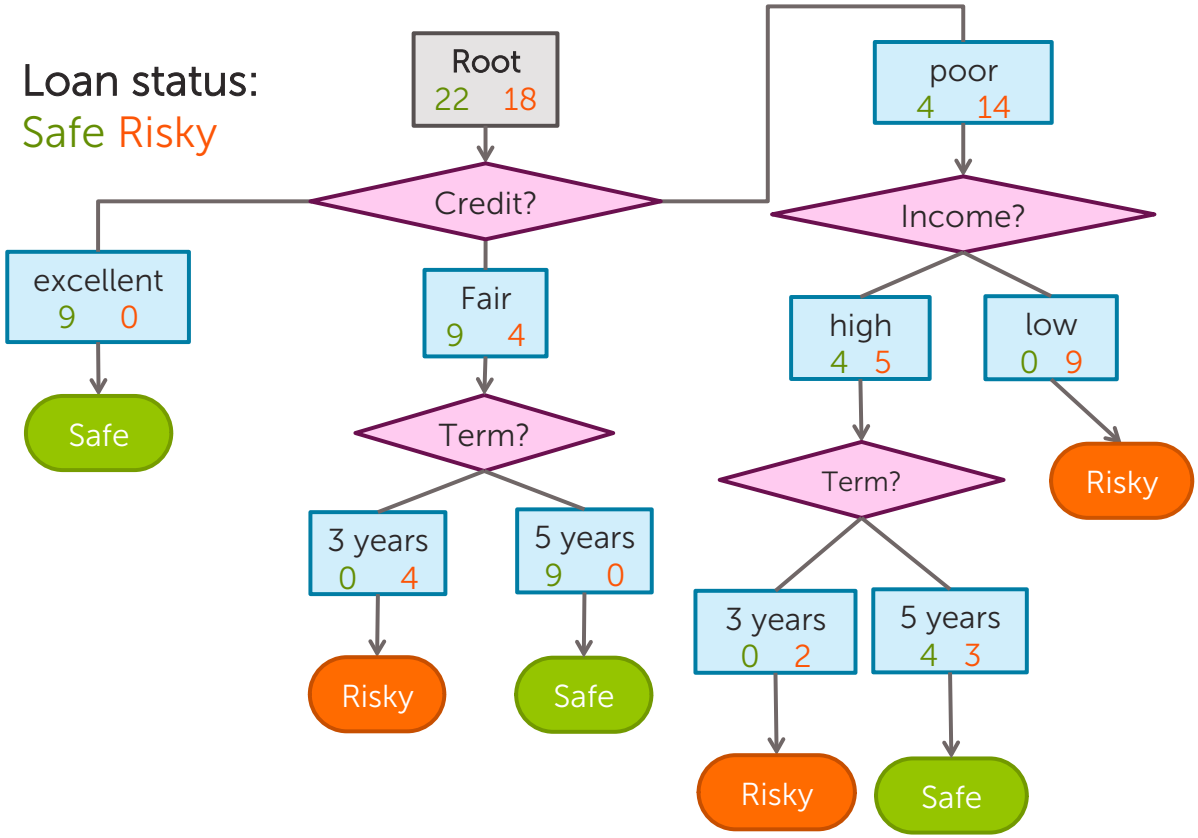
Final decision tree



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Simple greedy decision tree learning

Pick best feature to split on

Learn decision stump with this split

For each leaf of decision stump, recurse

When do we stop???

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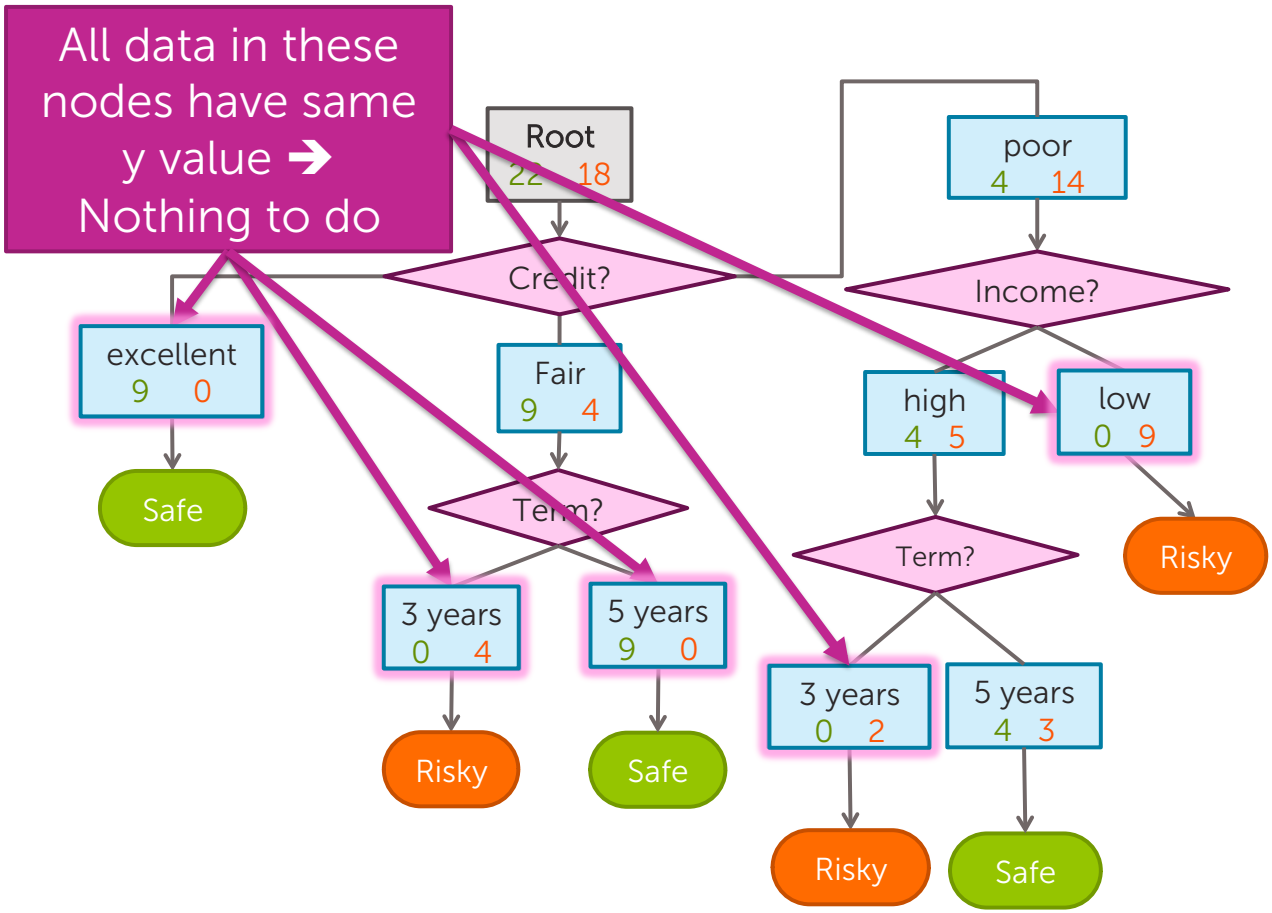
Stopping condition 1: All data agrees on y



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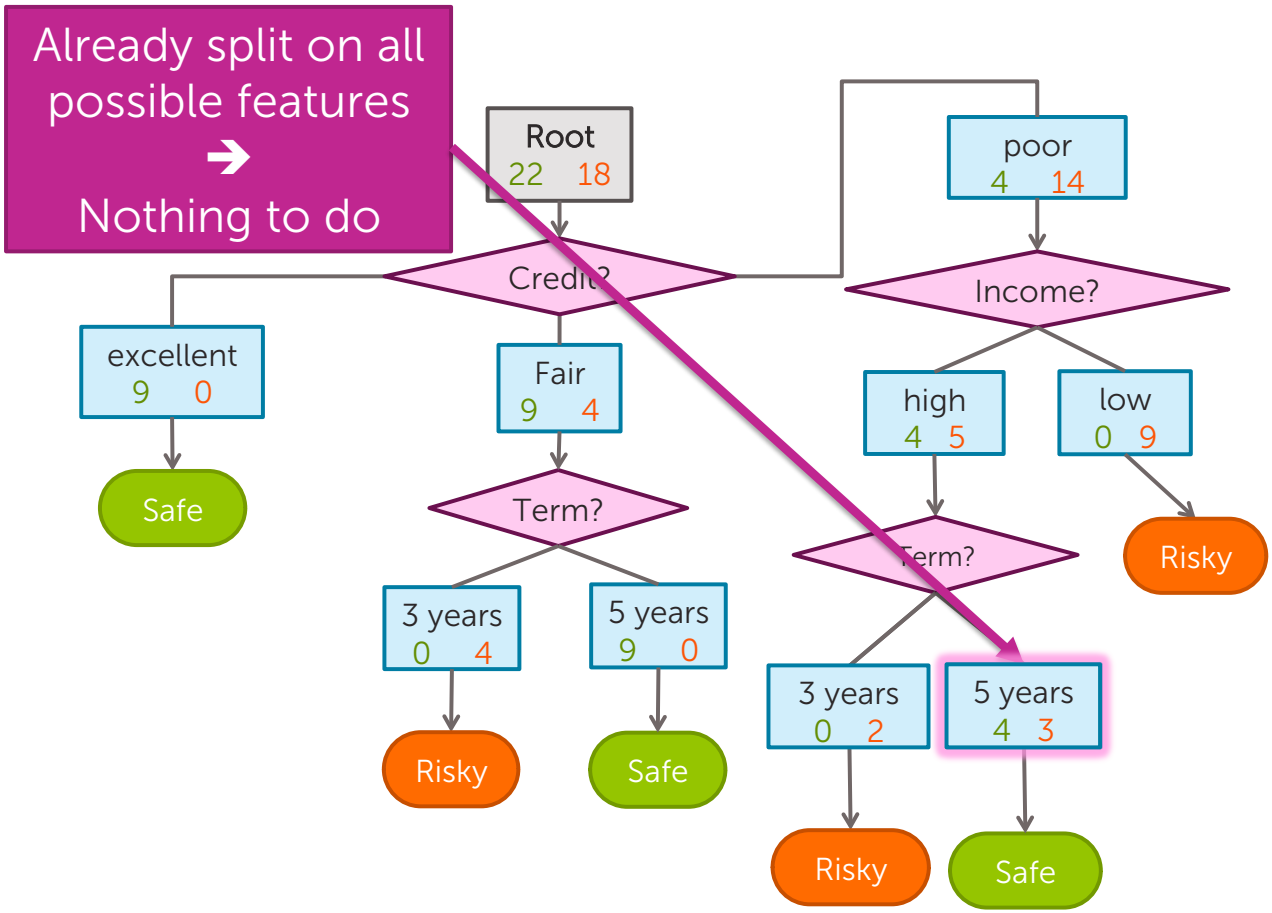
Stopping condition 2: Already split on all features



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Pick feature split leading to lowest classification error

Stopping conditions 1 & 2

Recursion

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Predictions with decision trees

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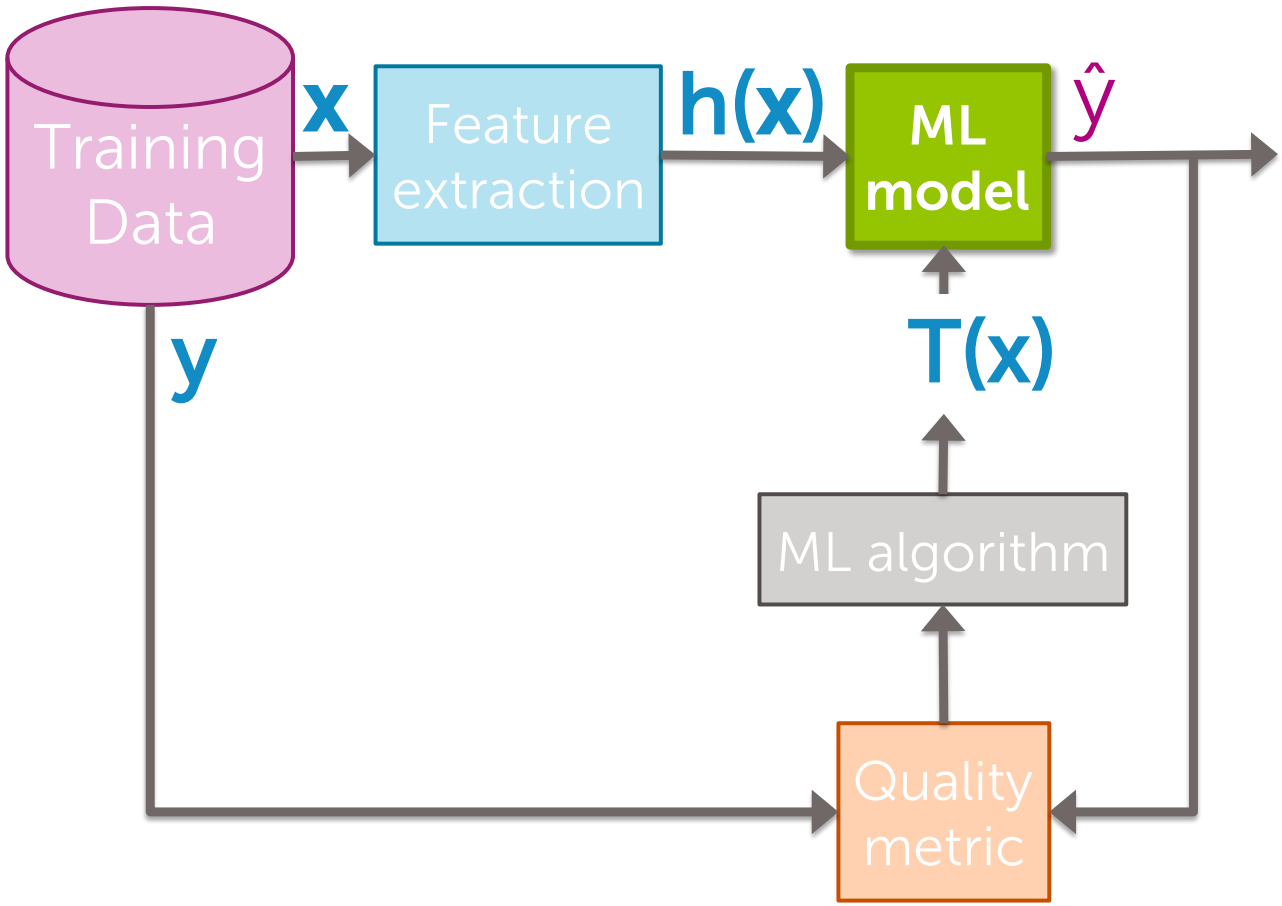

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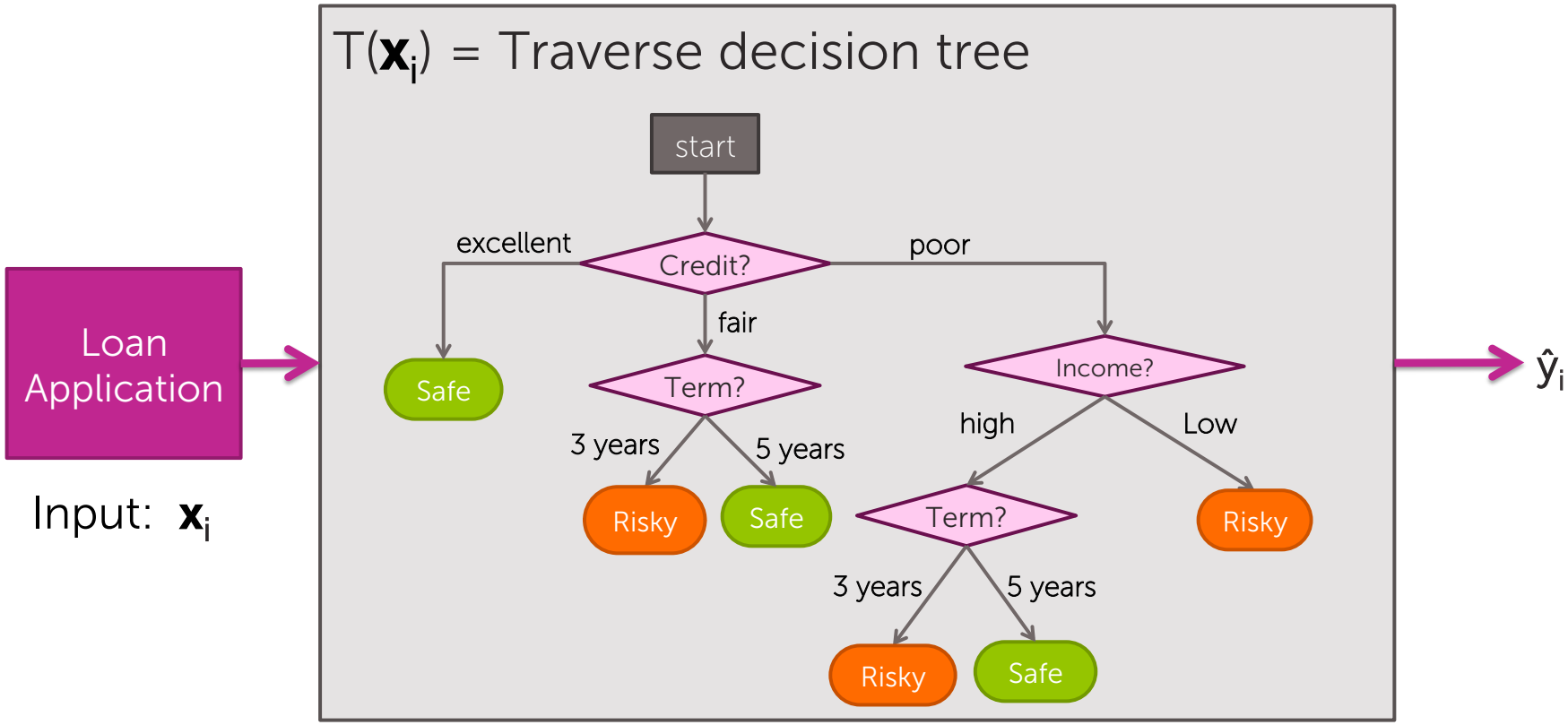
Decision tree model



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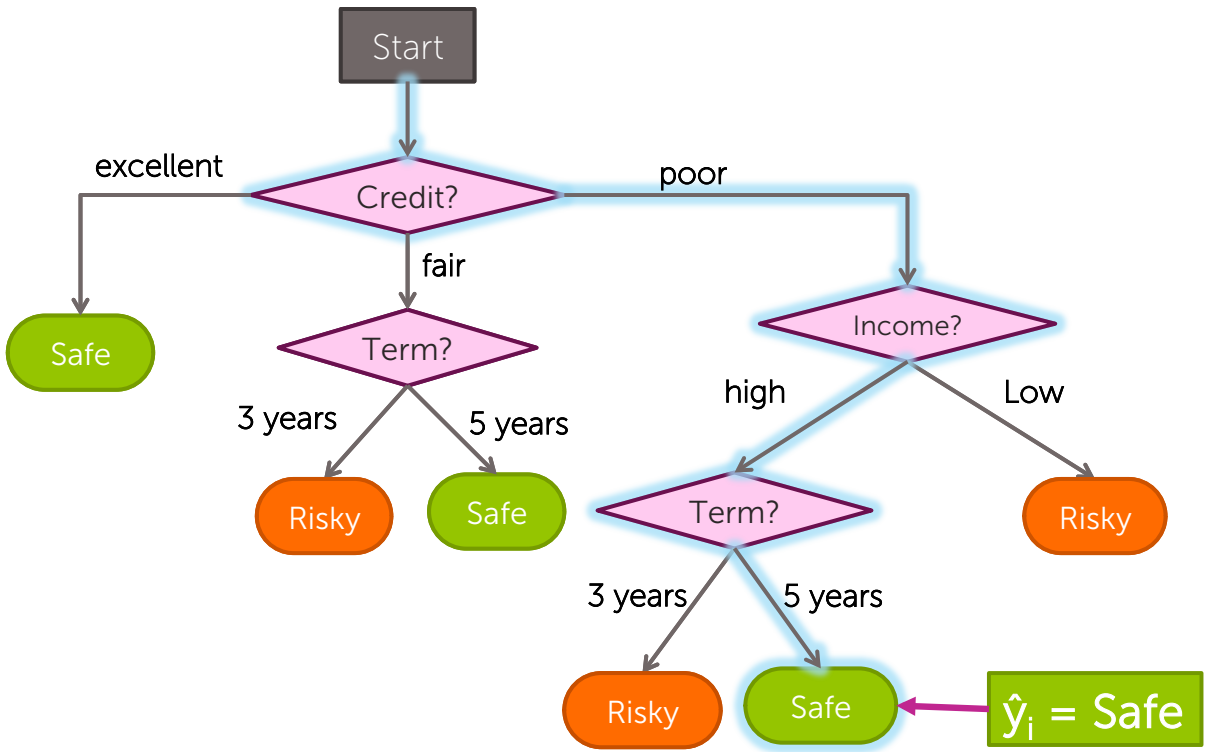
Traversing a decision tree

$\mathbf{x}_i$ = (Credit = poor, Income = high, Term = 5 years)



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Decision tree prediction algorithm

predict(tree_node, input)

- If current tree_node is a leaf:
  - return majority class of data points in leaf
- else:
  - next_note = child node of tree_node whose feature value agrees with input
  - return predict(next_note, input)

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Multiclass classification & predicting probabilities

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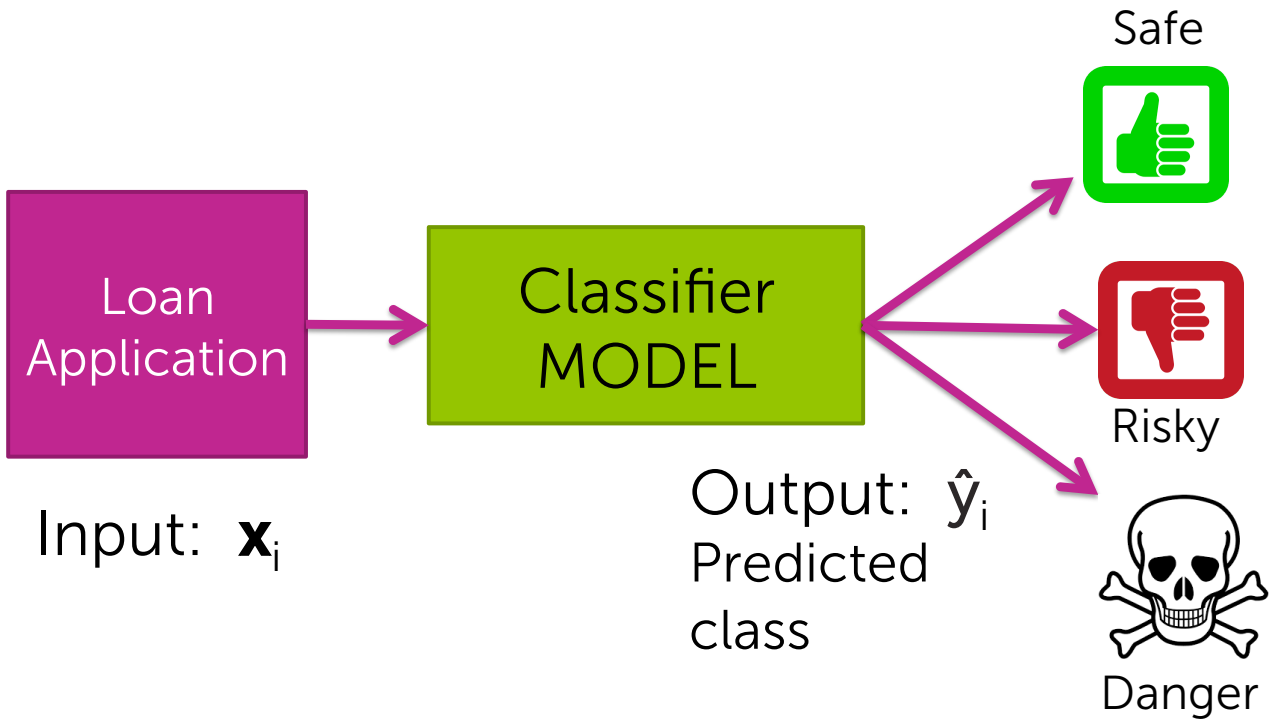
Multiclass prediction



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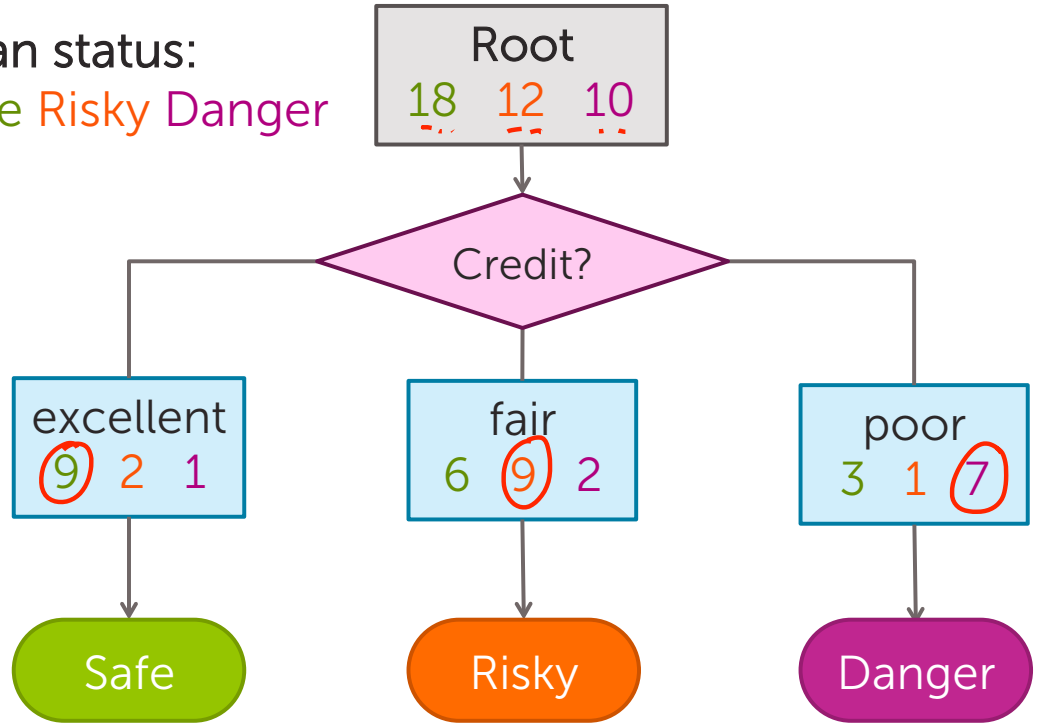
Multiclass decision stump

N = 40,
1 feature,
3 classes

Credit	y
excellent	safe
fair	risky
fair	safe
poor	danger
excellent	risky
fair	safe
poor	danger
poor	safe
fair	safe
...	...

Loan status:
Safe Risky Danger



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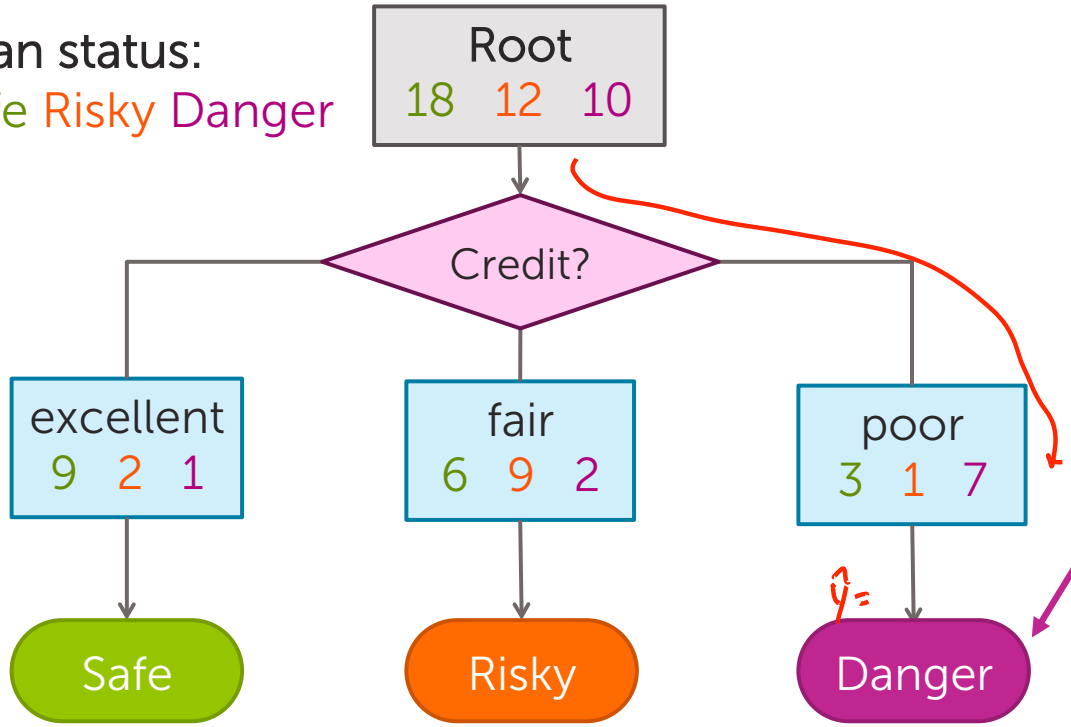
Predicting probabilities with decision trees

Loan status:
Safe Risky Danger



$$P(y = \text{danger} \mid \mathbf{x}) = \frac{7}{3 + 1 + 7} = 0.64$$

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Decision tree learning:
Real valued features

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How do we use real values inputs?

Income	Credit	Term	y
$105 K	excellent	3 yrs	Safe
$112 K	good	5 yrs	Risky
$73 K	fair	3 yrs	Safe
$69 K	excellent	5 yrs	Safe
$217 K	excellent	3 yrs	Risky
$120 K	good	5 yrs	Safe
$64 K	fair	3 yrs	Risky
$340 K	excellent	5 yrs	Safe
$60 K	good	3 yrs	Risky

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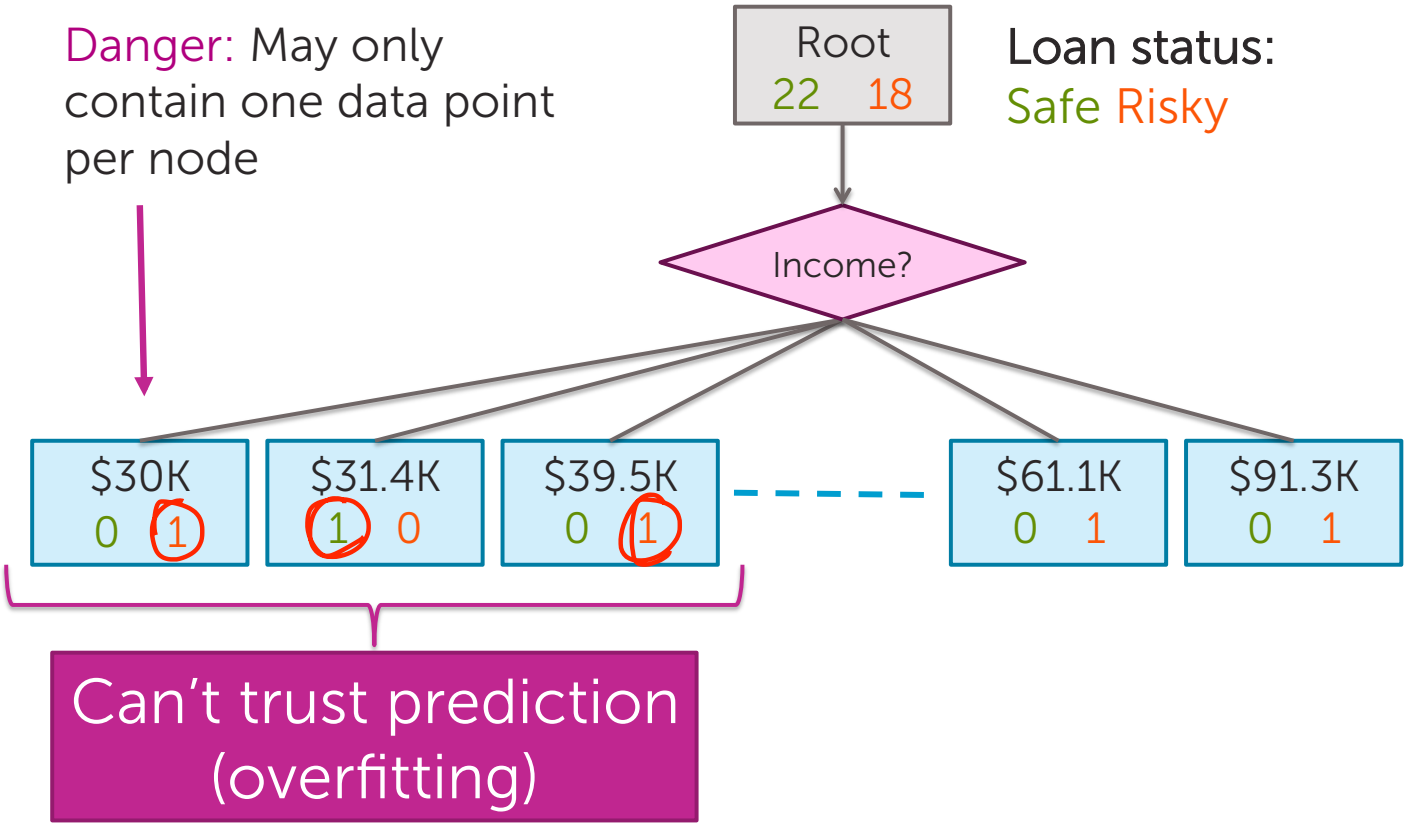
Split on each numeric value?

Danger: May only contain one data point per node



Can't trust prediction (overfitting)

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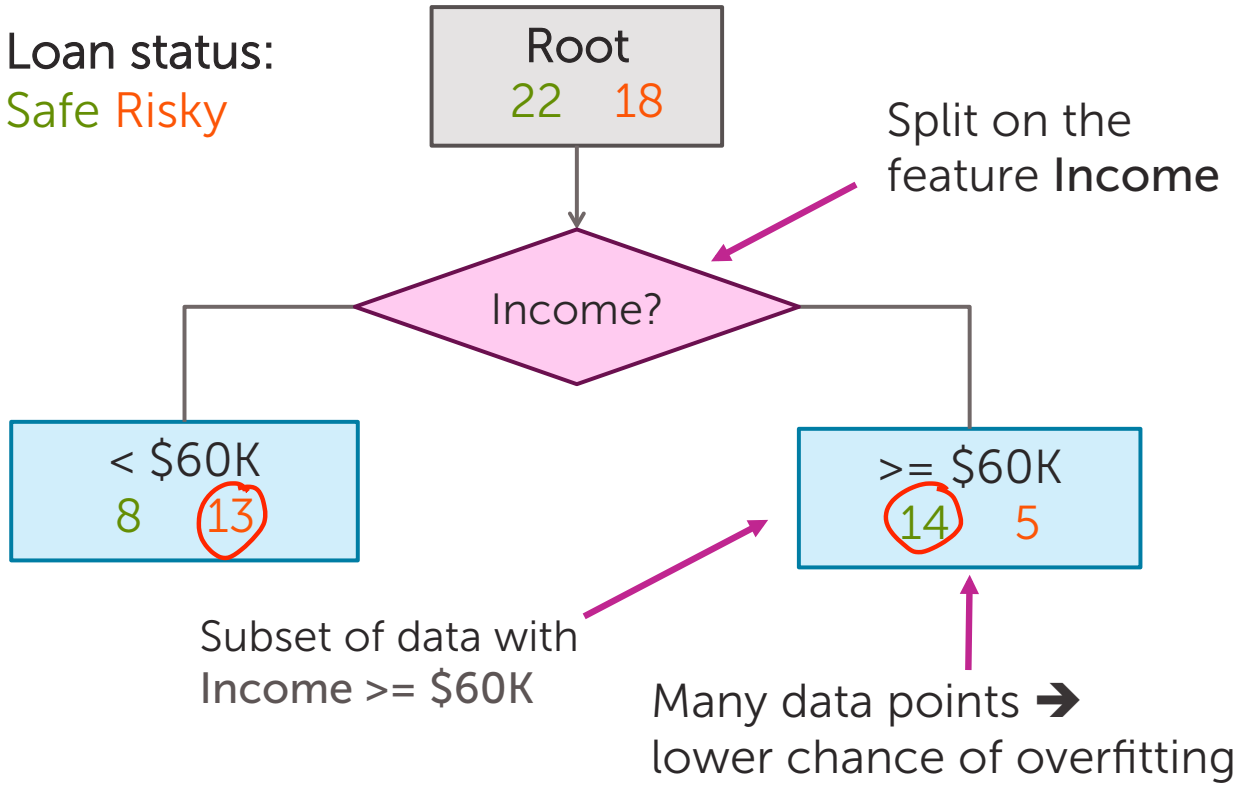
Alternative: Threshold split



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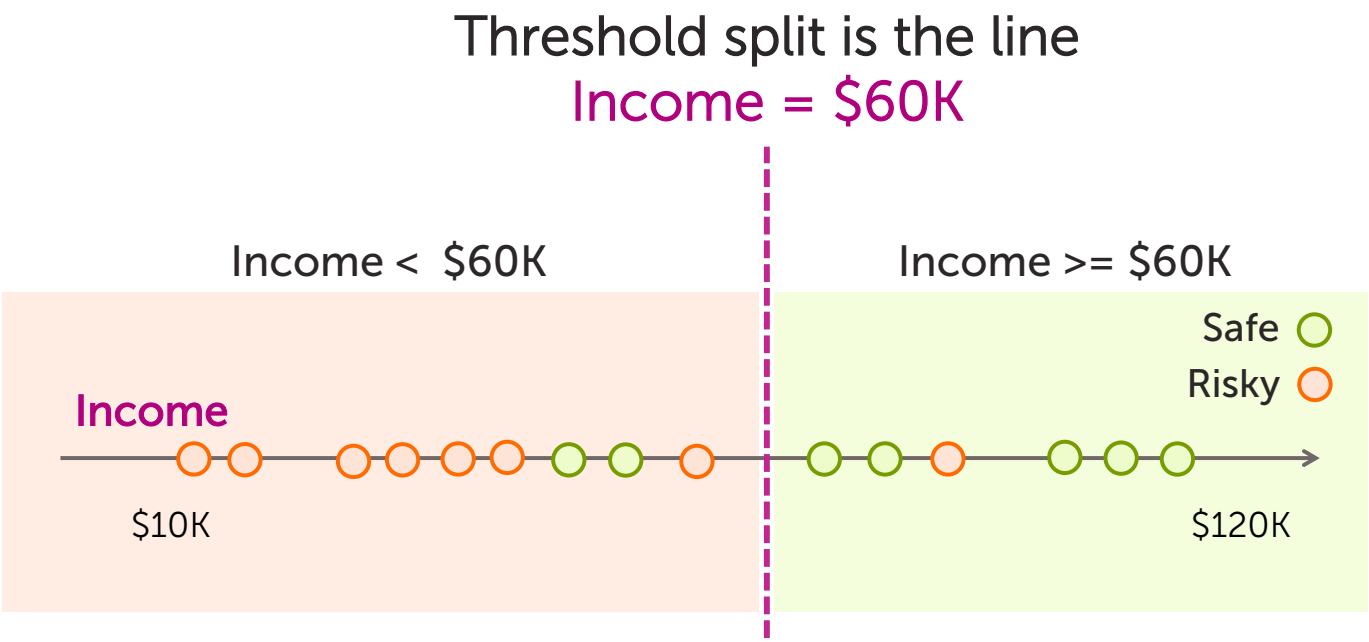
Threshold splits in 1-D



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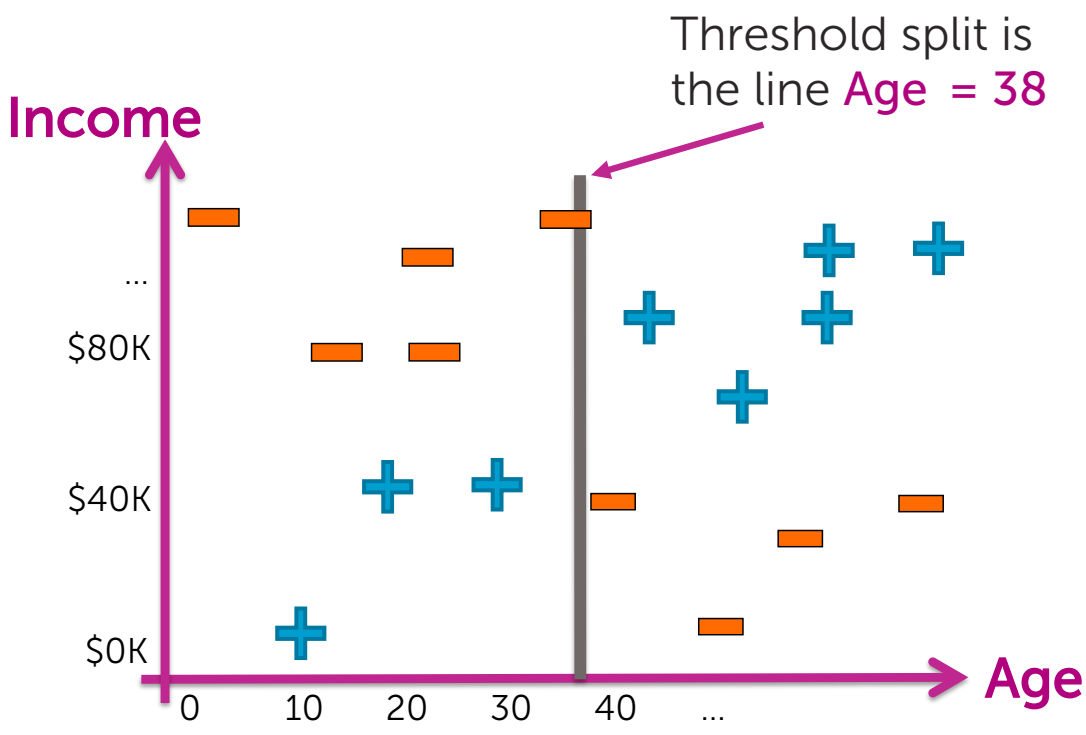
Visualizing the threshold split



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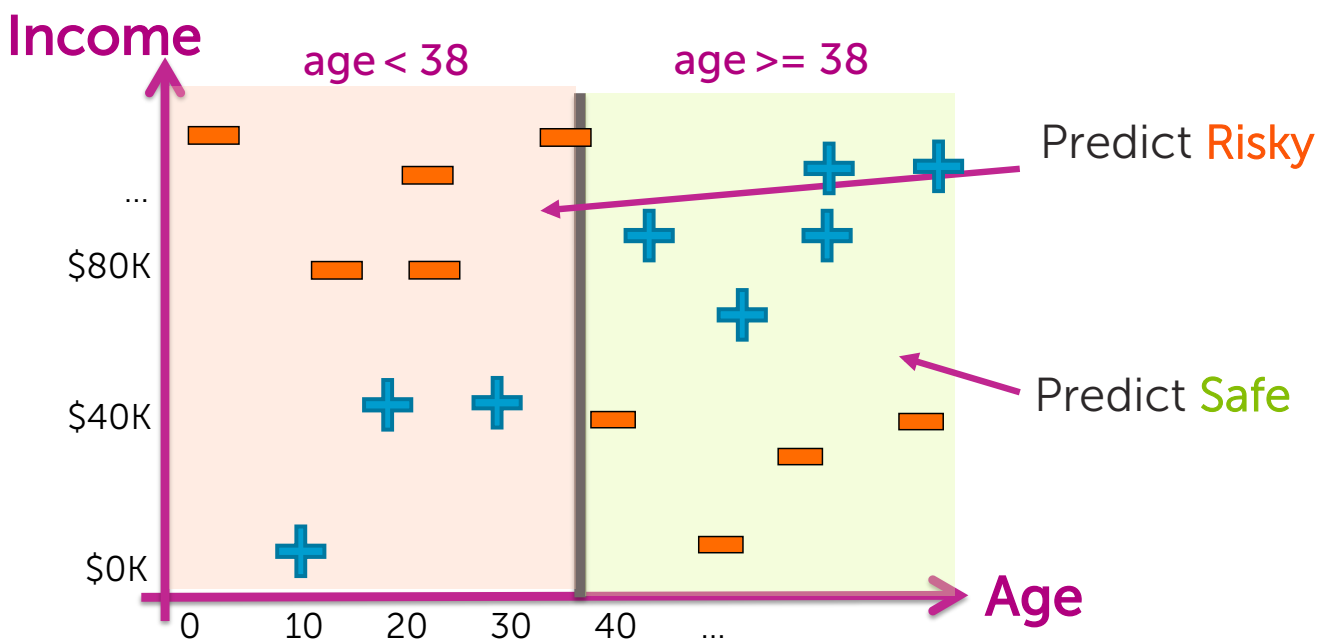
Split on Age >= 38



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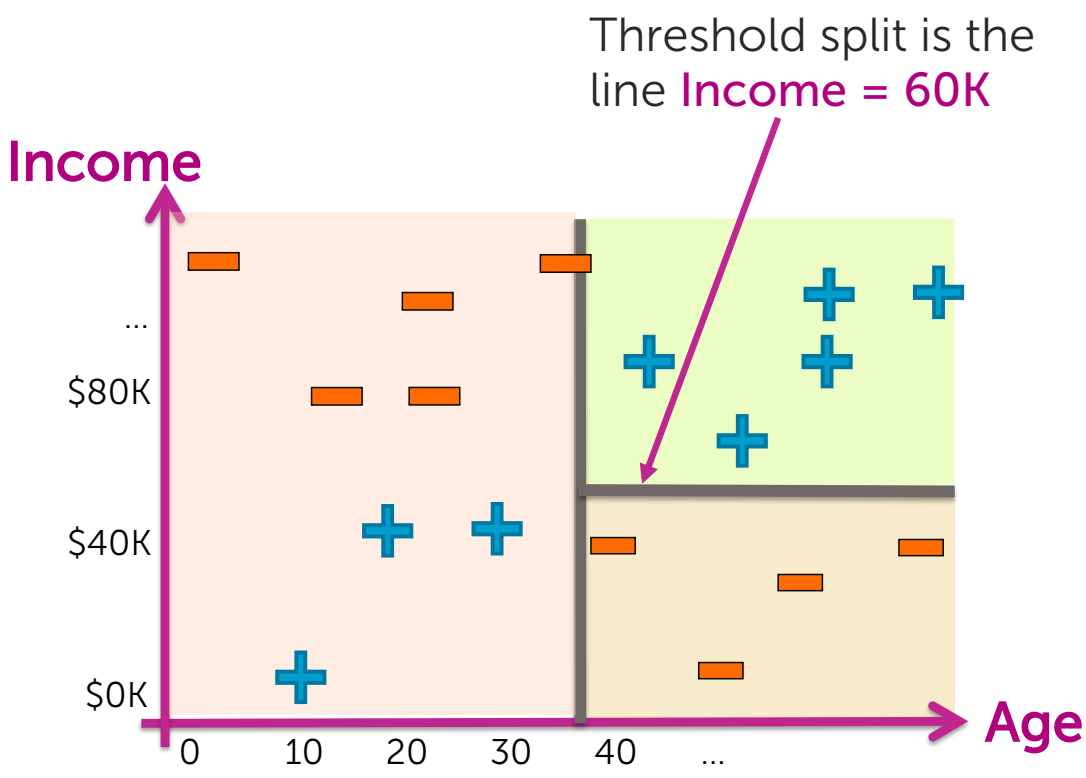
Depth 2: Split on Income >= $60K



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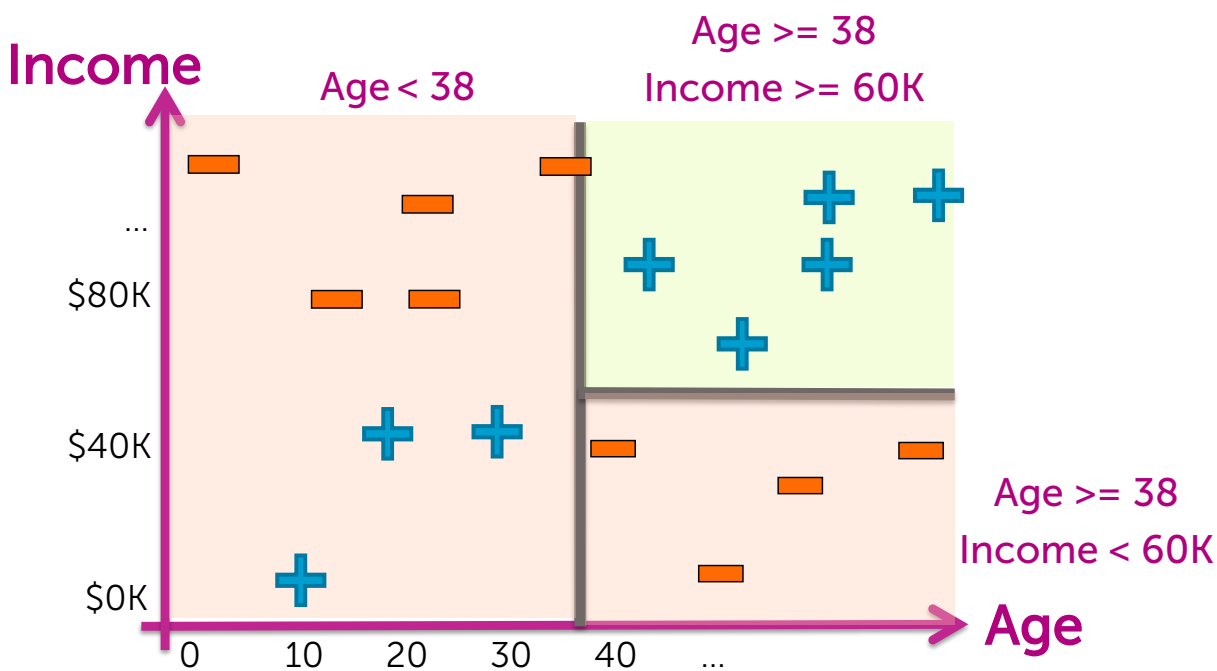
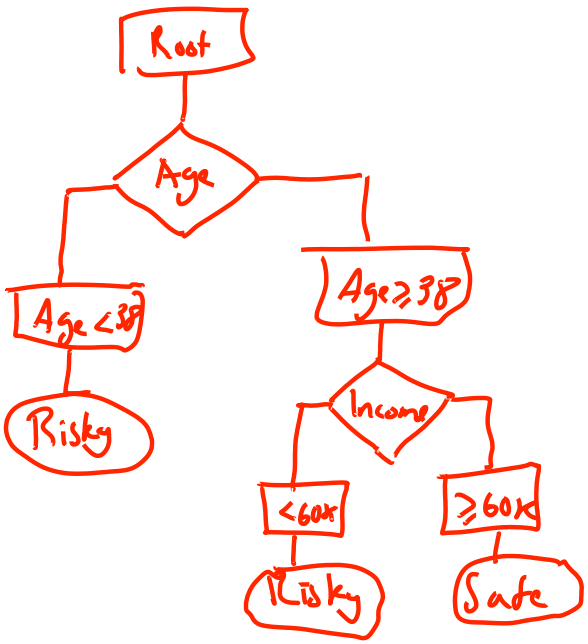
Each split partitions the 2-D space





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Finding the best threshold split

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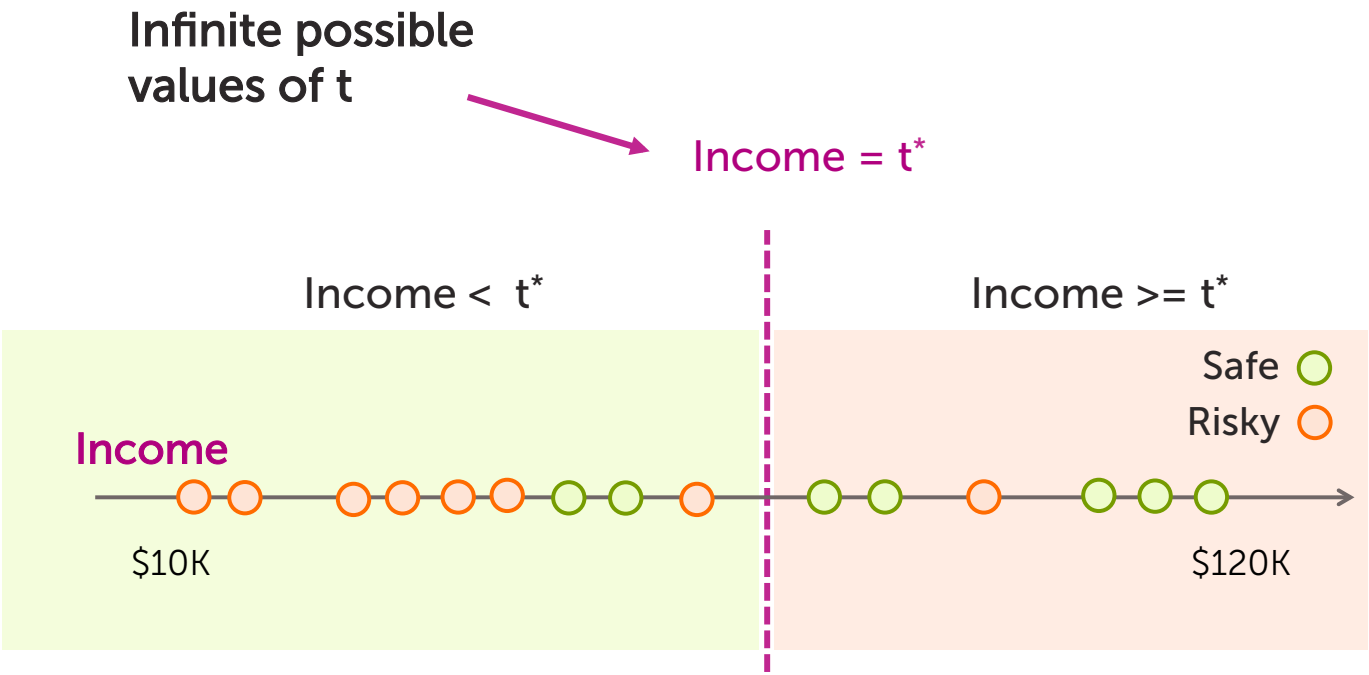
Finding the best threshold split



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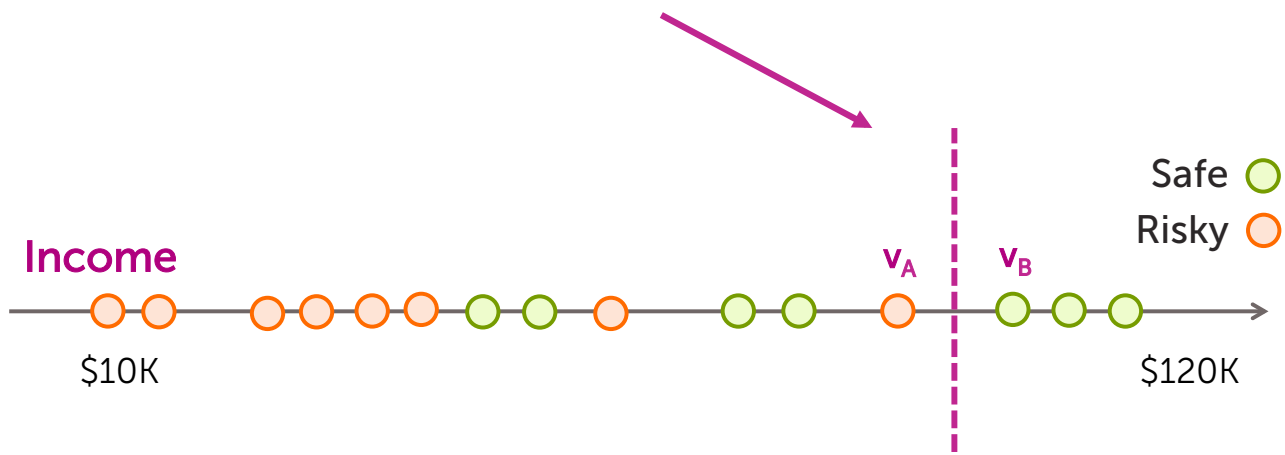
Consider a threshold between points

Same classification error for any threshold split between v_A and v_B



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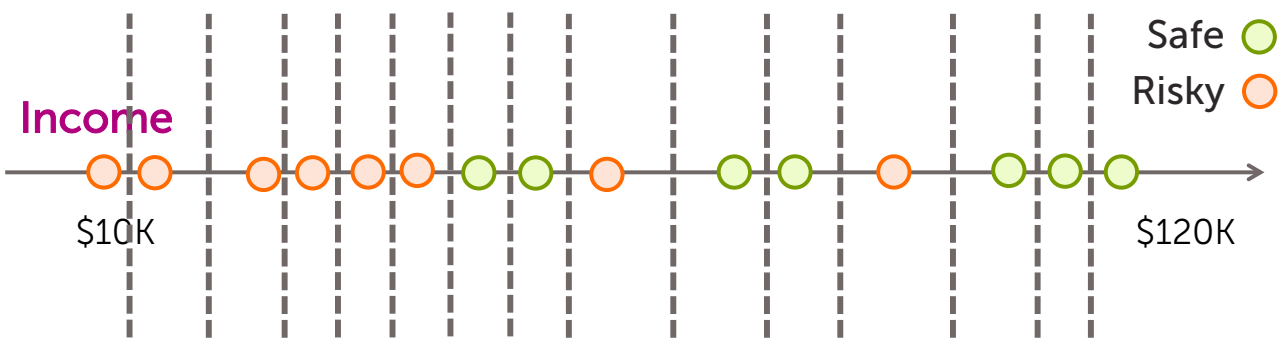
Only need to consider mid-points

Finite number of splits to consider



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Threshold split selection algorithm

income

- Step 1: Sort the values of a feature $h_j(\mathbf{x})$:

  Let $\{v_1, v_2, v_3, \ldots v_N\}$ denote sorted values
- Step 2:
  - For i = 1 … N-1
    - Consider split $t_i = (v_i + v_{i+1}) / 2$
    - Compute classification error for treshold split $h_j(\mathbf{x}) >= t_i$
  - Chose the t^* with the lowest classification error

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Decision trees vs logistic regression:
Example

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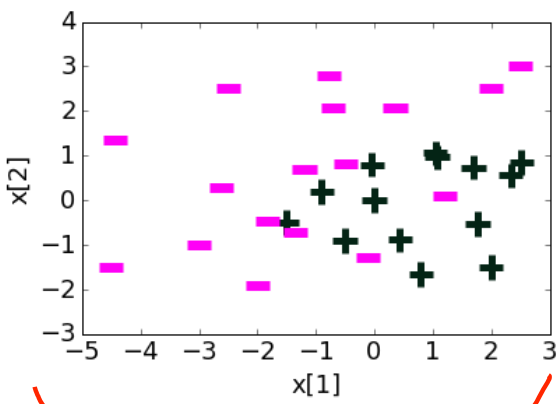
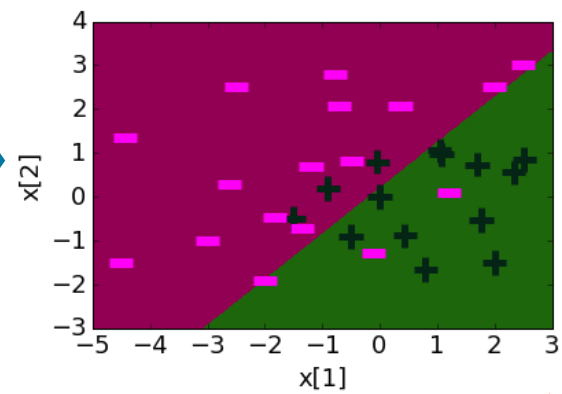
Logistic regression

Feature	Value	Weight Learned
$h_0(\mathbf{x})$	1	0.22
$h_1(\mathbf{x})$	$\mathbf{x}[1]$	1.12
$h_2(\mathbf{x})$	$\mathbf{x}[2]$	-1.07



data



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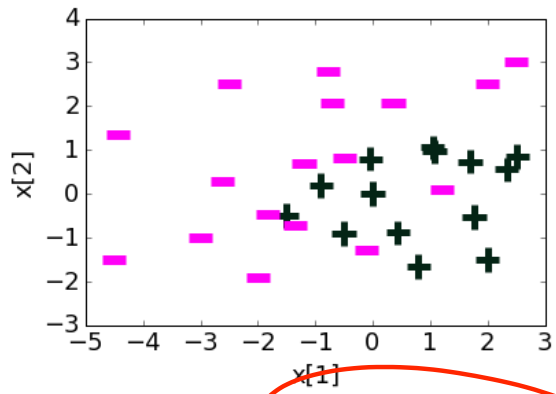
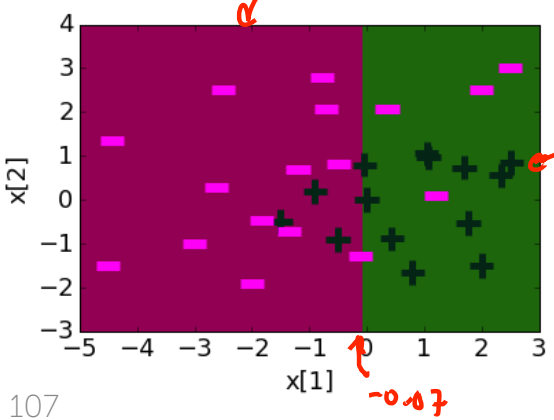
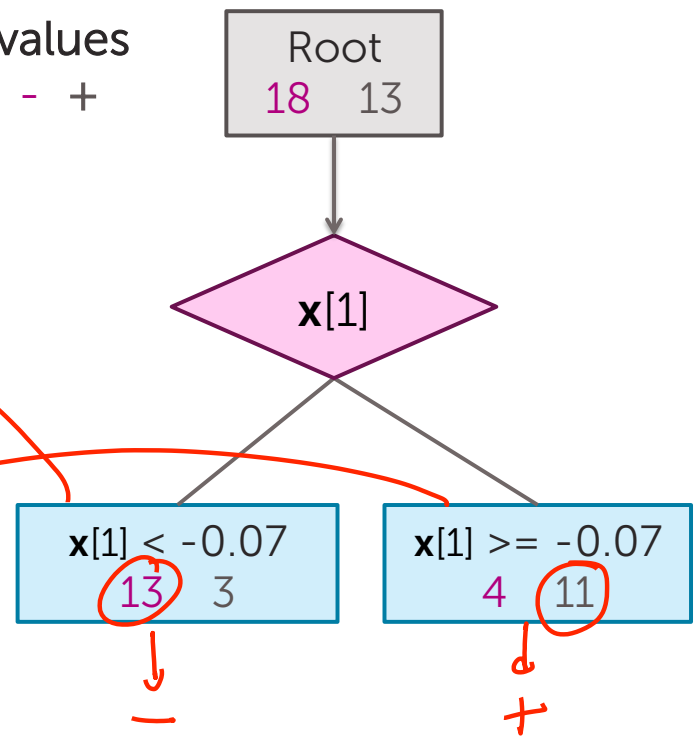
Depth 1: Split on **x**[1]





y values

- +



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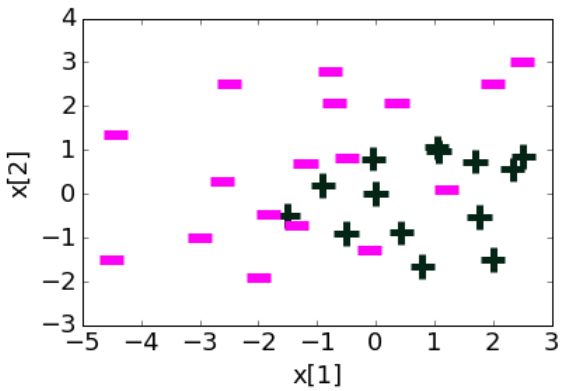
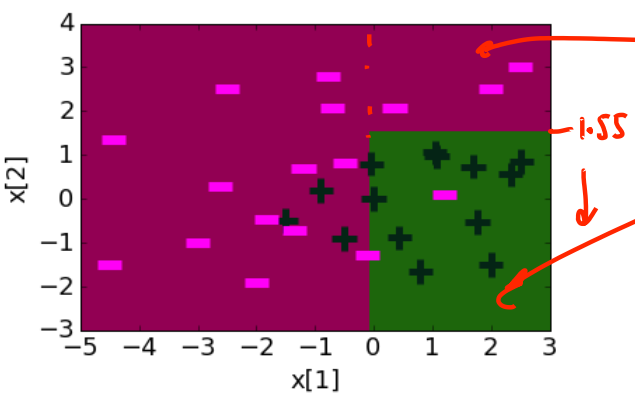
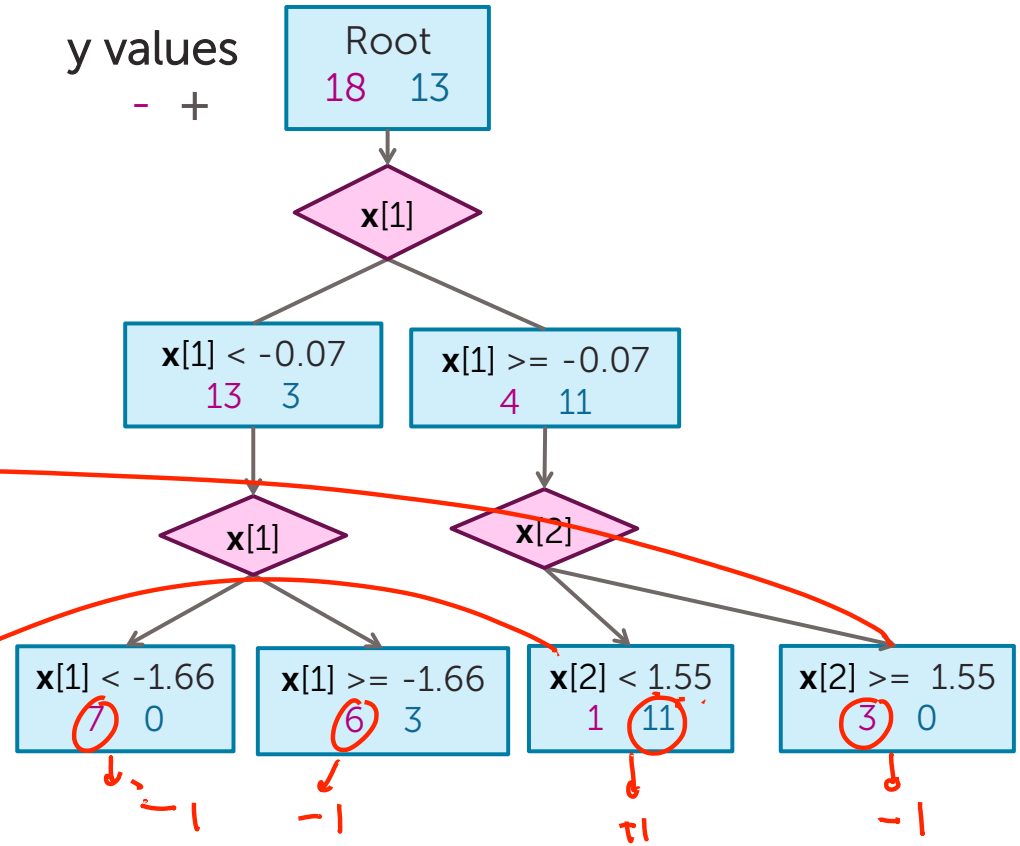
Depth 2







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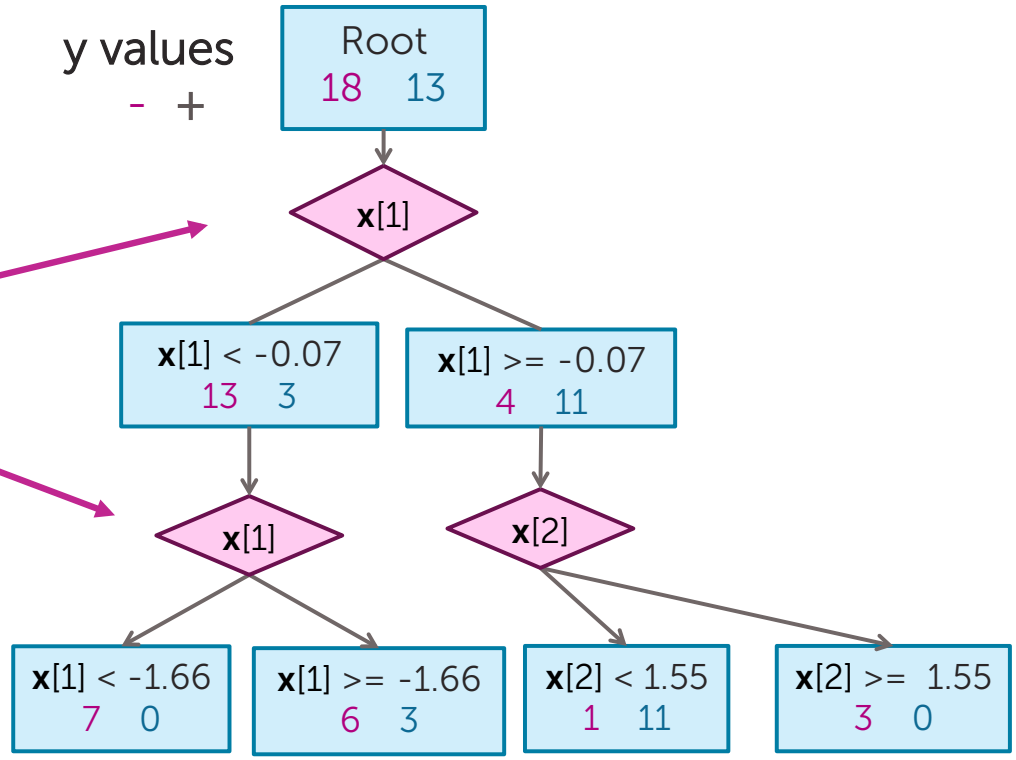
Threshold split caveat

For threshold splits, same feature can be used multiple times



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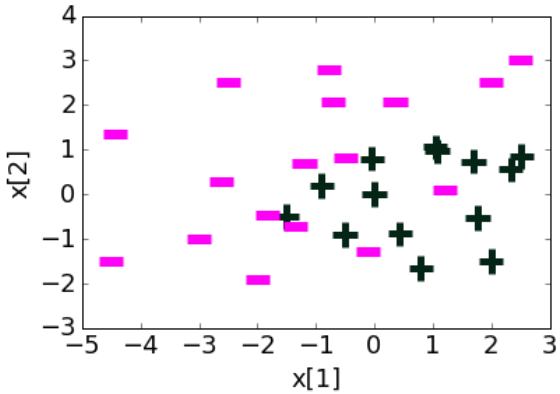
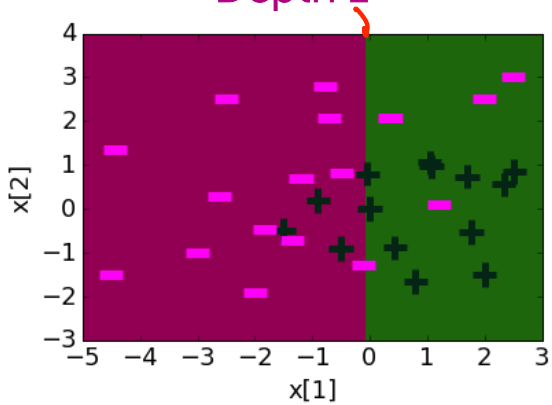
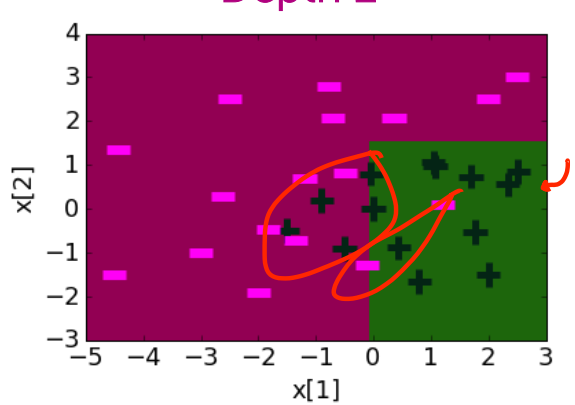
Decision boundaries



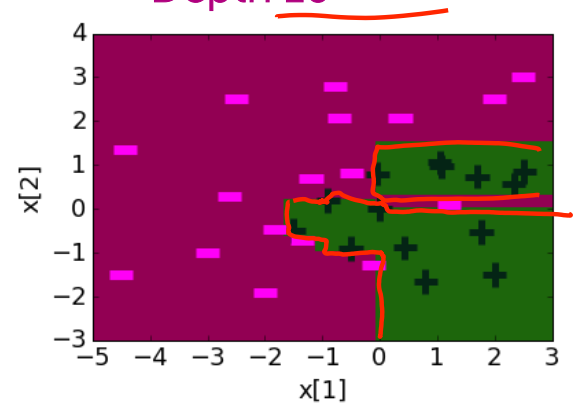
Depth 1



Depth 2



Depth 10



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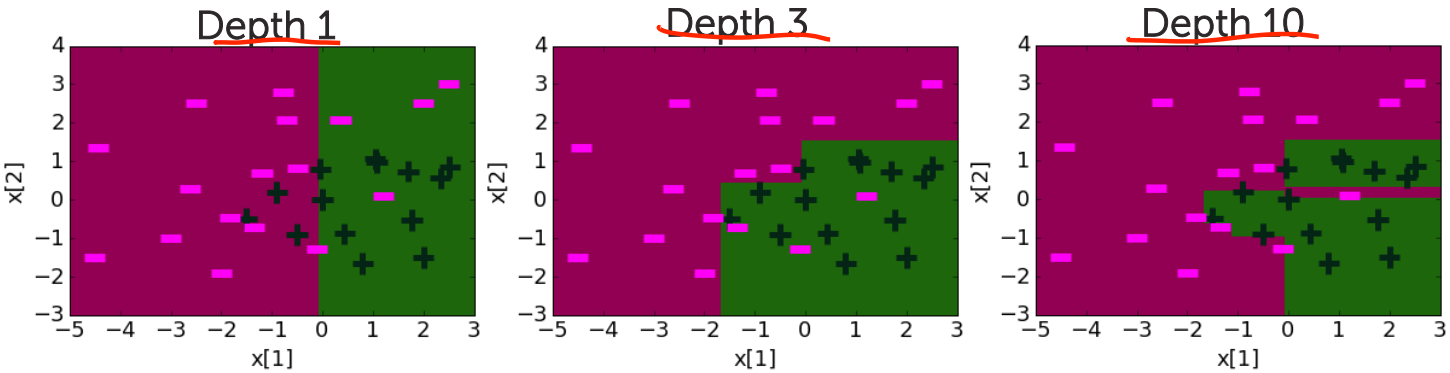
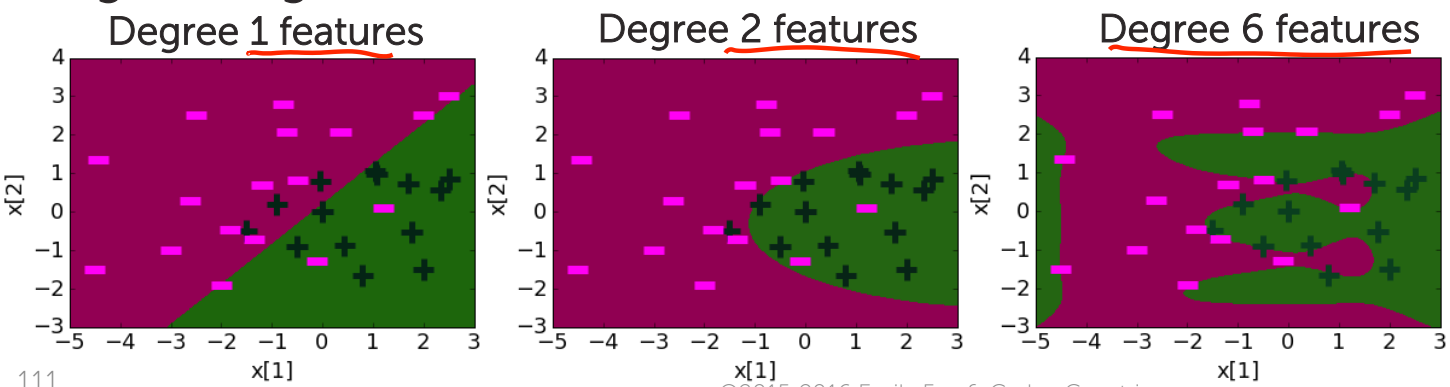
Comparing decision boundaries

Decision Tree



Logistic Regression



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Summary of decision trees

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What you can do now

- Define a decision tree classifier
- Interpret the output of a decision trees
- Learn a decision tree classifier using greedy algorithm
- Traverse a decision tree to make predictions
  - Majority class predictions
  - Probability predictions
  - Multiclass classification

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Thank you to Dr. Krishna Sridhar

Dr. Krishna Sridhar

Staff Data Scientist, Dato, Inc.

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